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Palaeogeography, Palaeoclimatology, Palaeoecology 213 (2004) 331–358

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Magnetostratigraphy and biostratigraphy of the Upper Triassic and lowermost Jurassic succession, St. Audrie's Bay, UK

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Received 23 May 2003; received in revised form 11 March 2004; accepted 16 July 2004

Abstract

The St. Audrie's Bay section in west Somerset comprises the uppermost Mercia Mudstone Group, the Penarth Group and the basal Lias Group and includes a candidate Global Stratotype Section and Point for the base of the Jurassic. The magnetostratigraphy has been evaluated through 122 m of this section at 147 stratigraphic levels, which range in age from mid-Norian to earliest Hettangian. In red dolomitic mudstones, the remanence is carried predominantly by haematite, whereas in non-red lithologies, it is mostly carried by magnetite. The mean virtual geomagnetic poles fall near the mean Upper Triassic and Lower Jurassic apparent polar wander track and display the start of the northeast-directed track typical of the Jurassic. The magnetostratigraphy comprises nine major magnetozones, five of normal polarity and four reversed, together with several minor magnetozones. In the Mercia Mudstone Group, the 68 m of the Twyning Mudstone Formation examined includes three major normal magnetozones (SA2n, SA3n and SA4n) and the Blue Anchor Formation has predominantly reversed polarity (SA4r) except at its top, in the Williton Member. The Penarth Group and basal Lias Group have predominantly normal polarity (SA5 to SA6n) but short reversals occur within the Westbury Formation and at the base of the Lilstock Formation and the Lias Group (SA5r).

This magnetostratigraphy is a good match with that found in the upper part of the Newark Supergroup succession in the eastern USA. The distinctive long reversal (SA4r) in the Blue Anchor Formation is equivalent to Newark Supergroup magnetozone interval E18r to E20r. Magnetozone SA5n, located mainly in the Penarth Group, probably equates with part of E22n and all of E23n in the Newark Supergroup. The reversed magnetozone, SA5r, at the base of the Lias Group, may correspond either with E23r in the Exeter Member (Passaic Formation) in the Newark Supergroup or with an undetected reverse polarity interval within the Newark Basin flood basalts. A change in the composition and diversity of terrestrial microfloras that occurs in the upper part of the Penarth Group at St. Audrie's Bay and elsewhere in the UK, is similar to that interpreted as marking the Triassic–Jurassic boundary in the Newark Supergroup. At St. Audrie's Bay, this change occurs c. 0.6 m below SA5r, within the Rhaetian, whereas in the Newark Supergroup, it occurs c. 20 m above the potentially equivalent E23r. Reconciliation of these disparities requires that either the microfloral changes are not

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doi:10.1016/j.palaeo.2004.07.018

synchronous between these locations or the change in the Newark Supergroup is time-equivalent with late Rhaetian conodont bearing strata. The correlation of marine and nonmarine Upper Triassic magnetostratigraphies is revaluated with the new data from St. Audrie's Bay, indicating the Twyning Mudstone and Blue Anchor formations are mid to late Alaunian (mid-Norian) in age. The Sevatian (late Norian) is represented by the Williton Member and the lower part of the Westbury Formation, but is incomplete because of disconformities at the base of the Williton Member and Penarth Group.
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Keywords: Magnetostratigraphy; Chronostratigraphy; Norian; Rhaetian; Hettangian; Triassic–Jurassic boundary

1. Introduction

Cliff and foreshore outcrops on the west Somerset coastline in south-west England provide almost continuous exposure of Upper Triassic and Lower Jurassic strata (Fig. 1). Palaeontological and lithological evidence shows the transition from continental playa–lacustrine environments, represented by the Twyning Mudstone Formation (Mercia Mudstone Group), to fully marine conditions represented by the Lias Group (Warrington and Ivimey-Cook, 1995; Figs. 2 and 3). The St. Audrie's Bay section is a candidate Global Stratotype Section and Point (GSSP; Remane, 2003) for the base of the Hettangian Stage and, therefore, the base of the Jurassic and, inter alia, the position of the Triassic–Jurassic boundary (Warrington et al., 1994). The lowest occurrence of ammonites of the genus *Psiloceras* has been used to mark this boundary in the British Isles (Cope et al., 1980; Warrington et al., 1980). However, Page and Bloos (1998) have shown in the St. Audrie's Bay area and elsewhere in England that this level is slightly below that of the appearance of *Psiloceras planorbis*, the index fossil of the lowest NW European Hettangian ammonite zone and subzone.

In the light of the importance of the St. Audrie's Bay section as a candidate GSSP, a detailed examination of the magnetostratigraphy and magnetic mineralogy of the Upper Triassic sediments, and the Triassic–Jurassic boundary strata, has been carried out by MWH and PEP. The biostratigraphy of the section has been reviewed by GW who has contributed the palynological study, which was carried out in conjunction with the survey of the Weston-super-Mare district by the British Geological Survey (Warrington, 1981; Whittaker and Green, 1983; Warrington and Whittaker, 1984).

2. Lithostratigraphy

Extensive exposures comprising nearly 200 m of Upper Triassic to Lower Jurassic strata occur on the west Somerset coast between Blue Ben [ST 1200 4390] and Doniford Bay [ST 0920 4325], on the southern side of the Bristol Channel Basin (Fig. 1). The exposures have been studied for more than 130 years, are well-documented (Richardson, 1911; Whittaker and Green, 1983; Warrington and Ivimey-Cook, 1995; Swift and Martill, 1999) and form part of a geological Site of Special Scientific Interest (SSSI) that extends from Blue Anchor in the west to Lilstock in the east (Fig. 1). The succession exposed comprises formations assigned to the Mercia Mudstone, Penarth and Lias groups (Figs. 2, 3 and 4) which occur widely at outcrop and beneath younger deposits in England and parts of South Wales, western Scotland and Northern Ireland and in the adjoining offshore areas (Warrington et al., 1980).

2.1. Mercia Mudstone Group

The greatest known thickness of the group in Somerset is 484 m, proved in the Burton Row Borehole, Brent Knoll (Whittaker and Green, 1983), 25 km to the east of St. Audrie's Bay (Fig. 1). Little of this succession is exposed, apart from the highest beds that are seen principally on the west Somerset coast, from St. Audrie's Bay eastwards to Blue Ben (Fig. 1). St. Audrie's Bay displays the uppermost 99 m in an unfaulted continuous section, where red-brown dolomitic mudstones of the Twyning Mudstone Formation (c. 68 m) are overlain by 31 m of the Blue Anchor Formation (Warrington and Ivimey-Cook, 1995). Farther east towards Blue Ben, a possibly older

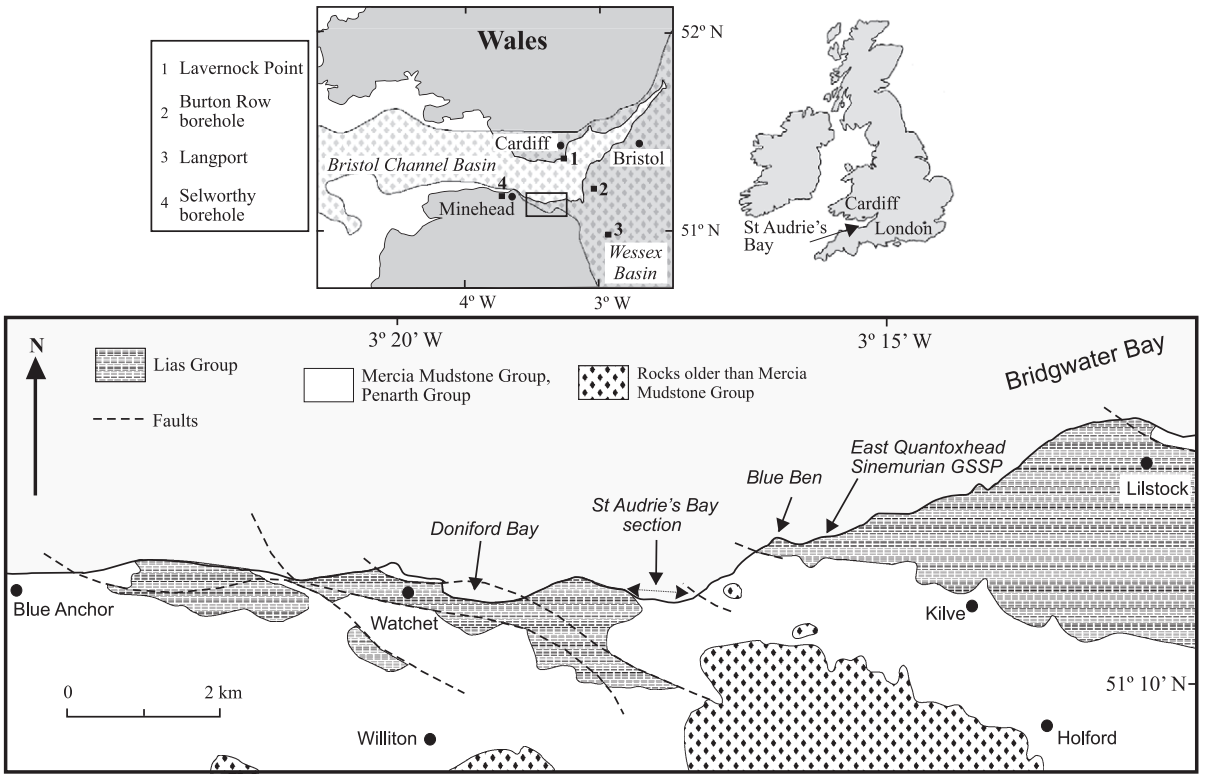


Fig. 1. Location and outline geology of the west Somerset coast around the St. Audrie's Bay section (based, with permission, on British Geological Survey 1:50,000 sheets 279—Weston-super-Mare (1980)—and 278—Minehead (1997)—which utilise Ordnance Survey mapping).

Twyning Mudstone Formation succession is interrupted by faulting.

2.1.1. Twyning Mudstone Formation

Barclay et al. (1997) introduced the formation name for the unit of the Mercia Mudstone Group underlying the Blue Anchor Formation in the Worcester Basin and neighbouring areas of England. It comprises red-brown dolomitic mudstones and siltstones with subordinate irregularly interbedded, green or greenish-grey mudstones and pale dolomites (Fig. 2), with occasional desiccation cracks, wave ripples and burrows (Talbot et al., 1994). The former presence of evaporites (sulphates) is indicated by dissolution voids and small collapse structures. The grey and green lithologies become more common upwards, and the base of the Blue Anchor Formation is placed arbitrarily at the top of the highest major red mudstone (Whittaker and Green, 1983; Warrington and Whittaker, 1984). The formation is thought to be of mixed floodplain and playa lake origin and to represent deposition in arid and semiarid continental environments of low relief, with possible soil development (Leslie et al., 1993; Talbot et al., 1994).

2.1.2. Blue Anchor Formation

This formation is divided into the Rydon and overlying Williton members (Mayall, 1981). The Rydon Member (29-m thick) comprises the "Tea Green Marls" and most of the overlying "Grey Marls" of older terminology (Richardson, 1911; Warrington et al., 1980; Warrington and Ivimey-Cook, 1995; Figs. 2 and 4). It predominantly includes grey, black, green and a few thin red-brown, dolomitic mudstones and dolostones; some laminated silty beds and mudcracks are also present. In the subsurface and at Blue Anchor (Fig. 1), the member contains gypsum nodules and beds, but at St. Audrie's Bay, solution features and collapse structures reflect the former presence of evaporite minerals (Mayall, 1979, 1981; Whittaker

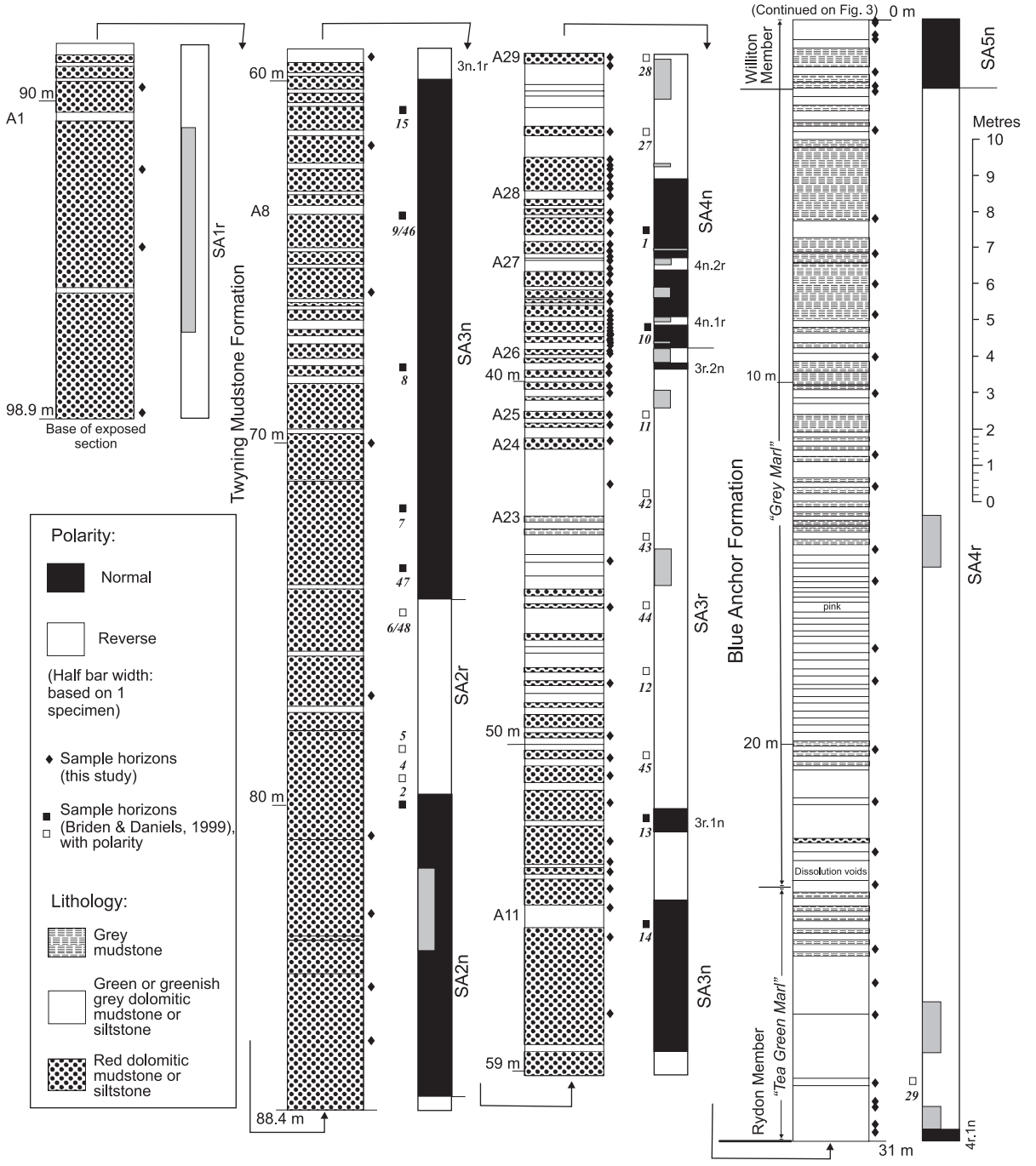


Fig. 2. Magnetostratigraphic sample horizons and polarity in the Mercia Mudstone Group, related to the log of Whittaker and Green (1983) slightly modified between 35 and 42 m, following detailed relogging of this interval.

and Green, 1983). The Williton Member comprises the highest beds of the former "Grey Marls" and is 2-m thick at St. Audrie's Bay, but thickens westwards (Mayall, 1981; Edwards, 1999). It has an irregular erosional base overlain by an intraformational conglomerate and penetrated by *Diplocraterion* burrows (Mayall, 1981). The member comprises grey silty shales with fine, flaser-bedded silts and sands and is overlain disconformably by the Westbury Formation (Penarth Group), above an erosion surface penetrated by *Diplocraterion* burrows.

2.2. Penarth Group

The Penarth Group (13-m thick) is exposed at the west side of St. Audrie's Bay and is subdivided into the Westbury Formation and the overlying Lilstock Formation (Fig. 3); the latter includes the Cotham and overlying Langport members.

The Westbury Formation consists of dark grey and black, shaley mudstones and subordinate silty, calcareous mudstones and grey, concretionary limestones. Also present are thin, wave-rippled sandstones, some of which are pebbly and contain remains of fish and other vertebrates, forming tempestite 'bone-beds' (Richardson, 1911; MacQuaker, 1994; Warrington and Ivimey-Cook, 1995; Swift and Martill, 1999). The top 5 cm of the formation at St. Audrie's Bay appears to be transitional in lithology to the overlying Cotham Member. This contrasts with the clearly erosional nature of this contact in South Wales and the Bristol region (Waters and Lawrence, 1987; Kellaway and Welch, 1993).

The Cotham Member of the Lilstock Formation (Fig. 3) comprises pale grey-green, calcareous mudstones with subordinate limestones, siltstones and sandstones, locally with mudcracks and stromatolitic developments (Mayall, 1983; Swift, 1999). A prominent horizon of disturbed sediment commonly occurs in the middle of the member, below a bedding surface penetrated by deep shrinkage cracks (Mayall, 1983; Hesselbo et al., 2002). This surface shows local evidence of erosion, although the deep shrinkage cracks are limited to the Bristol Channel region, and do not occur in the Cotham Member on the south Devon coast (Mayall, 1983). The Cotham Member formed in shallower water than the underlying Westbury Formation but contains a marine microflora and fauna and was deposited largely in marine environments (Mayall, 1979; Warrington, 1981; Swift, 1999). The overlying Langport Member consists of pale grey, commonly nodular but sometimes laminated, microspar and peloidal limestones that formed in shallow lagoons in warm, carbonate-rich marine water (Mayall, 1979; Warrington and Ivimey-Cook, 1995; Swift, 1999).

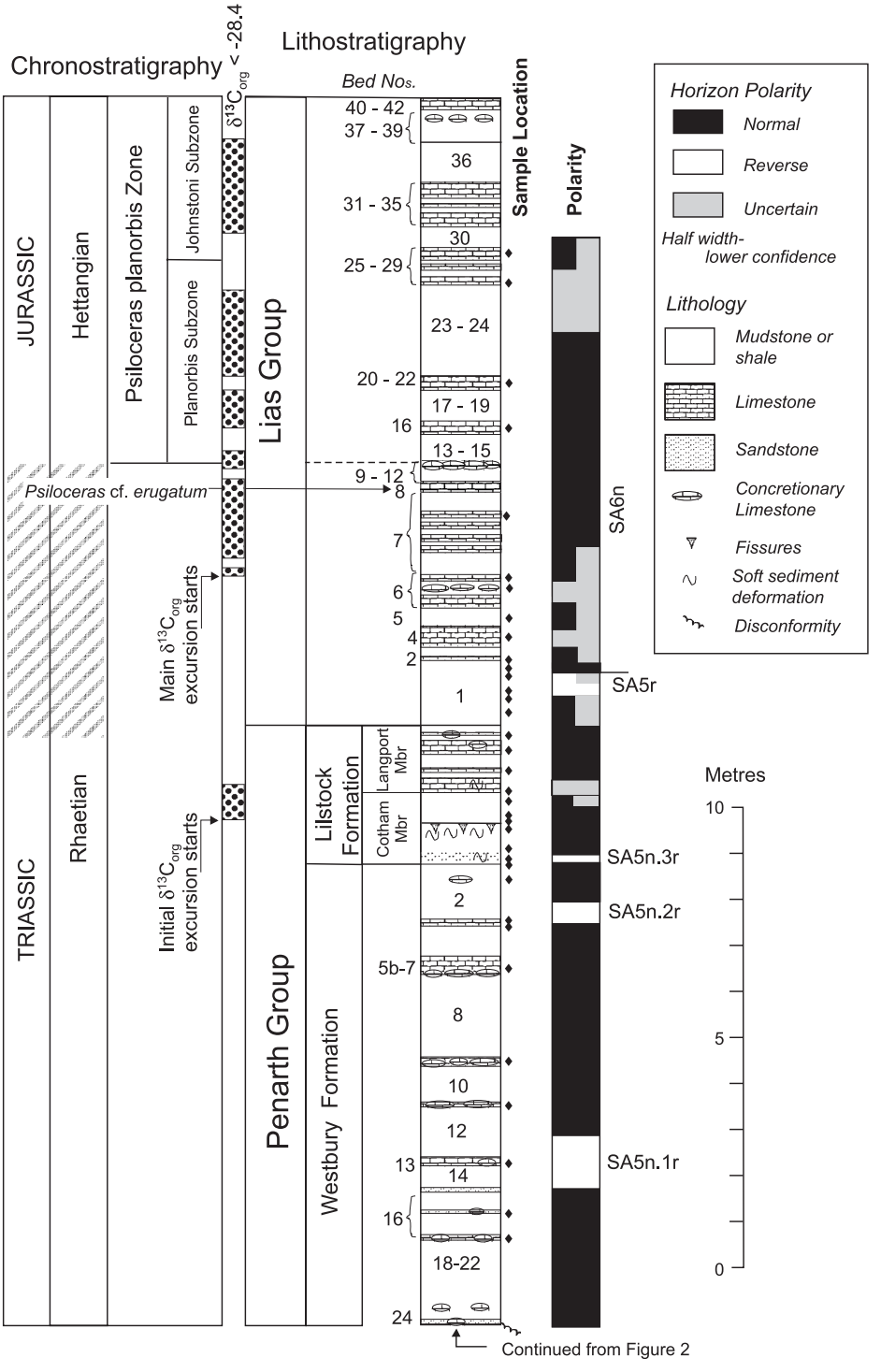
2.3. Lias Group

Cliffs and foreshore at and to the west of St. Audrie's Bay expose the lower 75 m of the Lias Group, including the Hettangian and part of the overlying Sinemurian (Warrington and Ivimey-Cook, 1995; Page and Bloos, 1998). The Lower Jurassic (Hettangian to Toarcian) succession in this region is 373-m thick in the Burton Row Borehole (Fig. 1; Whittaker and Green, 1983). The Lias Group consists of alternating limestones and shales with a marine biota.

3. Biostratigraphy

3.1. Palynology

In the terrestrial palynomorph record from St. Audrie's Bay (Fig. 5), circumpolles pollen, of cheirolepidacean origin, are a background constituent of the associations throughout the section studied. *Classopollis* spp. (*Corollina*; see Srivastava, 1976) are prominent in most assemblages, except in the lowest part of the Lias Group, where they are subordinate to *Gliscopollis meyeriana*. The latter occurs only in small numbers in assemblages from below the Langport Member, whereas *Granuloperculatipollis rudis* and *Geopollis zwolinskae* only occur below the Lilstock Formation. Most of the other taxa present here in the Twyning Mudstone Formation range upwards, through the Blue Anchor Formation, into the Penarth Group and, in some cases, the Lias Group. *Geopollis zwolinskae* is not definitely recorded above the Blue Anchor Formation and *Leptolepidites argenteaeformis* ranges only into the Westbury Formation. The highest records of *Ovalipollis pseudoalatus*, *Tsugaepollenites*? *pseudomassulae* and definite *Vesicaspora fuscus* are in the Cotham


Chronostratigraphy
Lithostratigraphy
< -28.4
δ13Corg
Bed Nos.
Sample Location
Polarity
JURASSIC
TRIASSIC
Hettangian
Rhaetian
Psiloceras planorbis Zone
Johnstoni Subzone
Planorbis Subzone
Lias Group
Penarth Group
Lilstock Formation
Langport Mbr
Cotham Mbr
Westbury Formation
Psiloceras cf. erugatum
Main δ13Corg excursion starts
Initial δ13Corg excursion starts
40 - 42
37 - 39
36
31 - 35
30
25 - 29
23 - 24
20 - 22
17 - 19
16
13 - 15
9 - 12
8
7
6
5
4
2
1
2
5b-7
8
10
12
13
14
16
18-22
24
SA6n
SA5r
SA5n.3r
SA5n.2r
SA5n.1r
Horizon Polarity
Normal
Reverse
Uncertain
Half width-lower confidence
Lithology
Mudstone or shale
Limestone
Sandstone
Concretionary Limestone
Fissures
Soft sediment deformation
Disconformity
Metres
10
5
0
Continued from Figure 2

Member, and *Acanthotriletes varius* extends into the Langport Member. *Rhaetipollis germanicus* ranges into the basal (pre-planorbis) beds of the Lias Group (Fig. 5) and has been recorded from the lower part of the Planorbis Subzone (Fisher and Dunay, 1981). A possible occurrence in the basal Hettangian has been noted in the present study (Fig. 5).

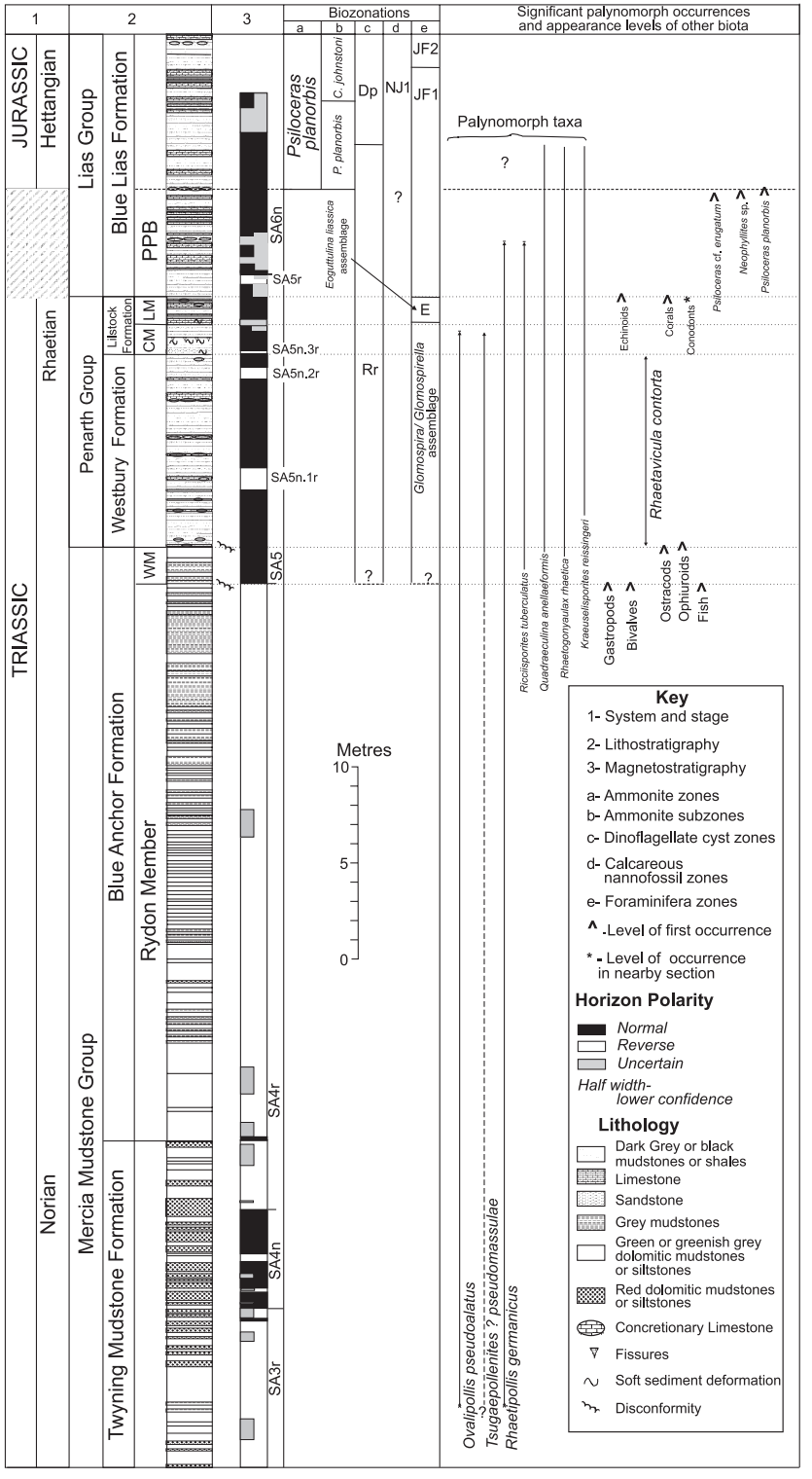
Progressive diversification of the microfloras appears to commence in the upper part of the Rydon Member of the Blue Anchor Formation and persists upwards through that formation and the succeeding Penarth Group. Many of the taxa, which contribute to this diversification, are species of trilete spore genera such as *Acanthotriletes*, *Carnisporites*, *Convolutispora*, *Deltoidospora*, *Kraeuselisporites*, *Microreticulatisporites*, *Perinosporites* and *Zebrasporites* (Fig. 5). Most taxa in these diverse associations have not been recorded above the Langport Member in this section. Exceptions include *Rhaetipollis germanicus*, *Ricciisporites tuberculatus*, which persists in small numbers into the pre-planorbis beds, and *Quadraeculina anellaeformis* and *Kraeuselisporites reissingeri*, which appear in the Williton Member and the Westbury Formation, respectively, and persist into the Jurassic (Fig. 5). An abrupt reduction in microfloral diversity occurs between the highest Langport Member assemblage and the lowest from the Lias Group, with circumpolles, principally *Gliscopollis meyeriana*, being dominant above the base of the Lias Group where they are associated with small numbers of *Kraeuselisporites reissingeri*, *Chasmatosporites*, non-taeniate bisaccate pollen and sporadic examples of *R. germanicus* and *R. tuberculatus*, the last definite occurrences of which are nearly 3 m below the appearance of the ammonite *Psiloceras planorbis* (Figs. 4 and 5).

The palynological record from St. Audrie's Bay (Fig. 5) is supplemented by sections at Lilstock (Warrington, British Geological Survey unpublished records), boreholes at Selworthy (Warrington et al., 1995; Edwards, 1999), Burton Row (Warrington, British Geological Survey unpublished records) and near Langport (Warrington et al., 1986), and the Lavernock section, South Wales (Orbell, 1973; Warrington in Waters and Lawrence, 1987). The record from Selworthy confirms the upwards increase in miospore diversity through the Blue Anchor Formation observed at St. Audrie's Bay. In addition, it shows a decline (from 10–20% to 1–5%) in the abundance of *Granuloperculatipollis rudis* at the top of the Rydon Member, which is less well displayed at St. Audrie's Bay.

At St. Audrie's Bay, the dinoflagellate cyst *Rhaetogonyaulax rhaetica* first appears in the topmost part of the Williton Member and is abundant in the Penarth Group below the top of the deformed horizon in the Cotham Member (Fig. 5). The first appearance of *R. rhaetica* is further below the base of the Westbury Formation in the Selworthy borehole, paralleling the westward thickening of the Williton Member (Edwards, 1999). The peak abundance of *R. rhaetica* is in the upper part of the Westbury Formation and lower part of the Cotham Member, as in other nearby sections (e.g., Warrington et al., 1995; Warrington, 1997; Waters and Lawrence, 1987) and elsewhere in England. A sharp reduction in the abundance of *R. rhaetica* occurs above the deformed horizon in the Cotham Member, just prior to the initial organic carbon isotope excursion at St. Audrie's Bay (Hesselbo et al., 2002; Fig. 3).

Dinoflagellate cysts (*Dapcodinium priscum* and *Rhaetogonyaulax rhaetica*) are scarce in organic-walled microplankton associations from the Lias Group but, as in other sections (Warrington, 1981), a greater variety of acritarchs is present than in the Penarth Group (Fig. 5). The upper limit of the Rr dinoflagellate cyst biozone of Woollam and Riding (1983) was modified by Powell (1992) and Riding and Thomas (1992) who placed it at the last occurrence of *R. rhaetica*, which they considered to be at the base of the Jurassic. However, the St. Audrie's Bay record (Fig. 5) shows that this taxon occurs sporadically at levels within the Hettangian Planorbis Zone.

Fig. 3. Magnetostratigraphic sample horizons and polarity in the Penarth Group and lower Lias Group, related to the logs and bed numbers of Richardson (1911) for the Westbury Formation and Whittaker and Green (1983) for the Lilstock Formation and Lias Group. Intervals of $\delta^{13}C_{org}$ which are less than −28.4 (‰ PDB) are shown, which define an initial isotopic excursion in the Cotham Member and the main isotopic excursion starting in the pre-planorbis beds (Hesselbo et al., 2002). The hatching indicates the Rhaetian–Hettangian boundary interval between the last occurrence of conodonts and the base of the *Psiloceras planorbis* Chronozone.


1
2
3
Biozonations
a
b
c
d
e
Significant palynomorph occurrences and appearance levels of other biota
TRIASSIC
JURASSIC
Norian
Rhaetian
Hettangian
Lias Group
Blue Lias Formation
Penarth Group
Lilstock Formation
Westbury Formation
Mercia Mudstone Group
Blue Anchor Formation
Rydon Member
Twyning Mudstone Formation
PPB
CM
LM
WM
Psiloceras planorbis
C. johnstoni
P. planorbis
Dp
NJ1
JF2
JF1
E
Rr
Equituflina liassica assemblage
Glomospira/ Glomospirella assemblage
SA6n
SA5r
SA5n.3r
SA5n.2r
SA5n.1r
SA5
SA4r
SA4n
SA3r
Metres
10
5
0
Palynomorph taxa
Ovalipollis pseudoalatus
Tsugaepollenites ? pseudomassulae
Rhaetipollis germanicus
Riccisporites tuberculatus
Quadraeculina anellaeformis
Rhaetogonyaulax rhaetica
Kraeuselisporites reissingeri
Rhaetavicula contorta
Gastropods
Bivalves
Echinoids
Corals
Conodonts *
Ostracods
Ophiuroids
Fish
Psiloceras cf. erugatum
Neophyllites sp.
Psiloceras planorbis
Key
1- System and stage
2- Lithostratigraphy
3- Magnetostratigraphy
a- Ammonite zones
b- Ammonite subzones
c- Dinoflagellate cyst zones
d- Calcareous nannofossil zones
e- Foraminifera zones
^ -Level of first occurrence
* - Level of occurrence in nearby section
Horizon Polarity
Normal
Reverse
Uncertain
Half width- lower confidence
Lithology
Dark Grey or black mudstones or shales
Limestone
Sandstone
Grey mudstones
Green or greenish grey dolomitic mudstones or siltstones
Red dolomitic mudstones or siltstones
Concretionary Limestone
Fissures
Soft sediment deformation
Disconformity

3.2. Microfossils

Foraminifera from the Blue Anchor Formation, Westbury Formation and Cotham Member represent the *Glomospira/Glomospirella* Assemblage (Fig. 4) of Copestake (1989). The Langport Member has yielded foraminifera of the *Eoguttulina liassica* Assemblage of Copestake (1989), and present in the Lias Group are foraminifera indicative of the JF1 and JF2 biozones (Fig. 4).

Hamilton (1982) reported the coccolith *Schizosphaerella punctulata* from the Cotham Member at St. Audrie's Bay. This fossil indicates the NJ1 biozone of Bown (1998) that extends from the Rhaetian–Hettangian boundary to the Angulata Zone (upper Hettangian). However, the occurrence of *S. punctulata* in the Penarth Group here, and at Lavernock (South Wales; Crux, 1987), has not been confirmed by later study which has restricted the NJ1 biozone to the Hettangian, based on sections from the Austrian Alps (Bown and Cooper, 1998). Ostracods recorded by Lord and Boomer (1990) from the Penarth Group and basal Lias Group, and by Swift (2003) from the Langport Member are indicative of the *Ogmoconchella aspinata* biozone of Boomer (1991).

Conodonts, including *Chirodella verecunda* and *Prioniodina*?, occur 0.15 m below the top of the Langport Member at Lilstock (Fig. 1; Swift, 1995) and have also been recovered near Watchet (Andrew Swift, personal communication to GW, 2001). In Nottinghamshire, these conodonts occur in association with *Misikella conformis* and *Misikella posthernsteini* in the Langport Member and pre-planorbis beds, respectively, both indicators of a Rhaetian age (Swift, 1995).

3.3. Ammonites and other macrofossils

Whittaker and Green (1983) recorded the lowest occurrence of ammonites (*Psiloceras planorbis*) within a shale unit (beds 13 to 15 of the Lias Group) at St. Audrie's Bay, and in bed 14 at Doniford Bay (Figs. 1 and 3). Ammonites were subsequently recovered from slightly lower (beds 8 and 9) at St. Audrie's Bay (Hodges, 1994), and a succession of ammonites has now been established there and at Doniford Bay (Page and Bloos, 1998; Bloos and Page, 2000). Bed 8 contains *Psiloceras* cf. *erugatum* and *?Psiloceras* sp., and cf. *Neophyllites* sp. occurs in the lower part of Bed 9 and *?Psiloceras* sp. cf. *P. planorbis* in its upper part (Figs. 3 and 4). Definite *P. planorbis* have not been recognised below the level identified by Whittaker and Green (1983). The Psiloceras planorbis Chronozone is divided into the Planorbis Subzone which, using the lowest occurrence of definite *P. planorbis*, comprises beds 13 to 28, and the Johnstoni Subzone, which comprises beds 29 to 42 (Figs. 3 and 4; Warrington and Ivimey-Cook, 1995). The succeeding Liasicus Zone, above the upper limit of the magnetostratigraphic study (Figs. 3 and 4), is defined by the appearance of *Waehneroceras* in bed 43 and extends to bed 79. The Jurassic succession up to beds in the Semicostatum Zone (Sinemurian) is also exposed on the west Somerset coast, which includes the GSSP site for the Sinemurian at East Quantoxhead (Bloos and Page, 2002; Fig. 1).

Other macrofossils are known from the upper 2 m of the Rydon Member and from the Williton Member (Mayall, 1981; Whittaker and Green, 1983; Warrington and Whittaker, 1984; Warrington and Ivimey-Cook, 1995; Fig. 4). The Westbury Formation has yielded a macrofauna dominated by bivalves but including (Fig. 4) an ophiuroid (*Aplocoma*). Mainly concentrated in the 'bone beds' are remains of fish and marine reptiles (Warrington and Ivimey-Cook, 1995). The macrofauna of the lowest c. 5 m of the Lias Group (beds 1 to 7 of Whittaker and Green, 1983) comprises bivalves and echinoid remains. The macrofauna of the Liasicus Zone is more diverse than that of the Planorbis Zone and includes, in addition to ammonites, a coral, bivalves, echinoid fragments and fish remains (Warrington and Ivimey-Cook, 1995).

Sections, such as that in New York Canyon (Nevada), that show a more complete Upper Triassic ammonite or conodont biostratigraphy than the

Fig. 4. Summary of uppermost Triassic and lowermost Jurassic biostratigraphy at St. Audrie's Bay, related to the lithostratigraphy and magnetostratigraphy of the section. The lower c. 50 m of the Twyning Mudstone Formation in Fig. 2 is unfossiliferous and is not shown. Abbreviations, see Fig. 5. The hatching indicates the Rhaetian–Hettangian boundary interval between the last occurrence of conodonts and the base of the *Psiloceras planorbis* Chronozone.

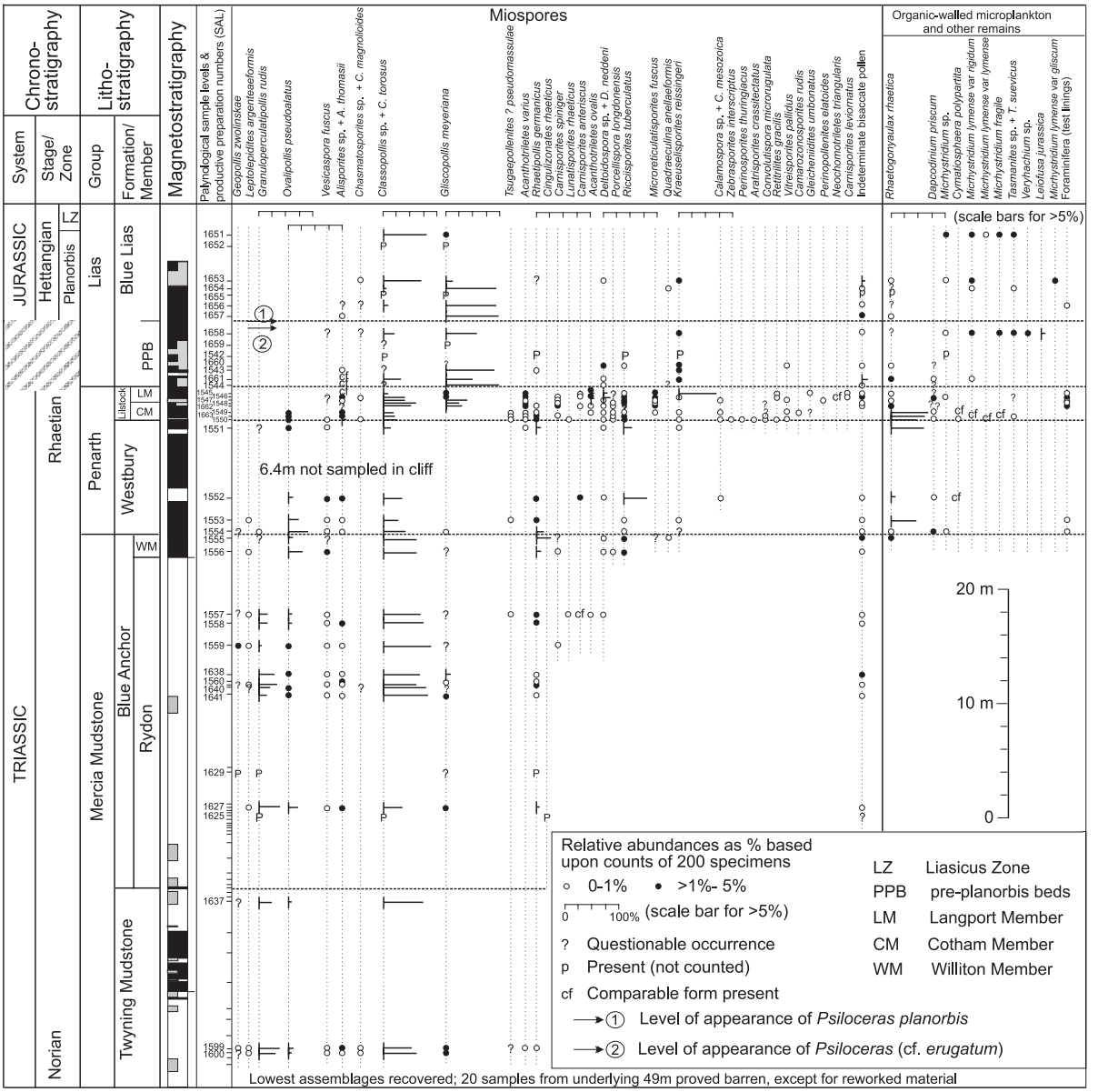


Fig. 5. Distribution and relative abundances of palynomorphs in the St. Audrie's Bay section (from G. Warrington: British Geological Survey internal reports PDL.72/190 (1972) and PDL.79/287 (1979)). Palynology preparations curated at the British Geological Survey (Keyworth) and are registered in the SAL series. The hatching indicates the Rhaetian–Hettangian boundary interval between the last occurrence of conodonts and the base of the *Psiloceras planorbis* Chronozone.

succession at St. Audrie's Bay, indicate that conodonts became extinct prior to the establishment of the first typical Jurassic ammonites such as *Psiloceras* and *Neophyllites* (Guex et al., 2003). Consequently, pending a decision on a GSSP for the Hettangian, the Triassic–Jurassic boundary interval at St. Audrie's Bay lies between the last occurrences of conodonts on the top of the Lilstock Formation and the occurrence of *Psiloceras planorbis* (Figs. 3, 4 and 5). The isotopic and floral changes indicated here and in

Hesselbo et al. (2002), below the Lias Group, therefore occur within the Rhaetian.

4. Palaeomagnetism

4.1. Methods

Palaeomagnetic samples were collected at 147 horizons in a continuous 122-m-thick succession from an easterly bounding fault in the Twyning Mudstone Formation (Fig. 1), to the Johnstoni Subzone in the Lias Group (Figs. 2 and 3). Conventional palaeomagnetic core plug samples (2–3 per horizon) and hand samples were used, orientated by a magnetic compass. The specimens were measured on a CCL GM400 3-axis cryogenic magnetometer (noise level~0.002 mA/m) which made six measurements of each specimen axis, from which the magnetisation variance was calculated. Demagnetisation was performed using a Magnetic Measurements Limited thermal demagnetiser and a MOLSPIN alternating field (AF) demagnetiser. Red specimens were typically demagnetised between 150 and 700 °C, in 50 °C increments, with 25 °C increments at higher temperatures. Non-red specimens were generally thermally demagnetised to 350 °C (250 °C for samples from the Penarth and Lias groups), followed by AF demagnetisation from 10 to 80 mT in 10 mT steps. This composite demagnetisation scheme is similar to that used by Yang et al. (1996) and was adopted because heating above ~250–350 °C induced changes in the mineralogy of non-red lithologies, which obscured the isolation of the Characteristic Remanent Magnetisation (ChRM). Susceptibility measurements were made after each heating step to monitor the mineralogical changes and to identify the susceptibility crisis. ChRM directions were isolated using principle component analysis as implemented in LINEFIND (Kent et al., 1983). Use has been made of both linear trajectory fits and great circle data in defining the palaeomagnetic behaviour. LINEFIND contains a statistical test to evaluate the significance of planarity in fitting great circles to the demagnetisation data and displays those that are significant.

Progressive Isothermal Remanent Magnetisation (IRM) up to 1T, was applied to a representative subset of specimens, to investigate the magnetic mineralogy, which was also studied by demagnetising a three-component orthogonal IRM (Lowrie, 1990). Magnetic fields of 1000, 300 and 100 mT were applied along the respective x, y and z axes of the red specimens, and 1000, 100 and 40 mT fields were applied along the corresponding axes of the non-red specimens. Hysteresis measurements (up to 1 T) were performed on a MOLSPIN "Nuvo" Vibrating Sample Magnetometer (VSM).

4.2. Palaeomagnetic behaviour

At low demagnetisation temperatures, the samples were commonly found to be contaminated by a steep positive, commonly northerly, magnetisation that was strongest in the non-red lithologies. This overprint is interpreted as a Brunhes-age magnetisation. Specimen behaviour during demagnetisation was classified into the following three types:

(a) **S-Type**: A ChRM could be defined, using three or more points on a straight line segment, mostly directed through the origin. In non-red lithologies, the ChRM was commonly defined between 10 and 80 mT (SA2.3, Fig. 6a) or 350 °C to 40 mT AF range (SA67.1, Fig. 6b). In red lithologies, the ChRM line-fits mostly used demagnetisation steps above 500 °C, but with some specimens, the high temperature components were conformable from c. 300 °C. Specimens were visually classified into three S-type groups (S_1, S_2 and S_3), based on the length and precision of the line fit, with S_1 having the highest precision (Fig. 6a and b) and S_3 the lowest. S-type behaviour is seen in 44% of the specimens measured (Fig. 7). The average excess standard deviation parameter (ρ, Kent et al., 1983) was 1.9 (average α_{95}=12°), indicating that the fitted lines average variance is slightly more than the magnetometer measured variance.

(b) **T-Type**: Specimens exhibited a great-circle path towards a down NE direction or up SW direction, commonly from a steep northerly positive direction. This type of behaviour is most evident in reversely magnetised specimens, but is also common in normally magnetised

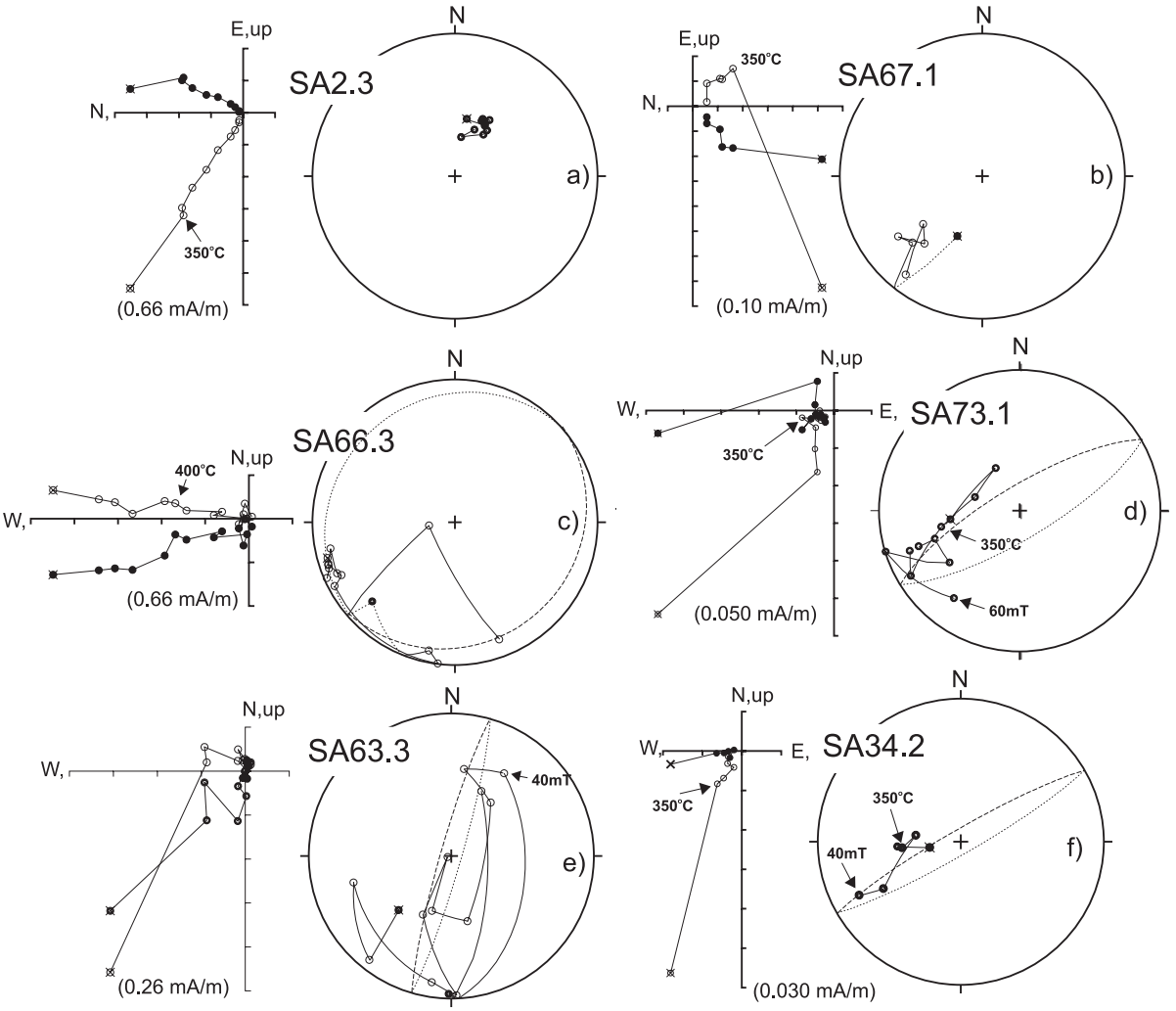


Fig. 6. Representative demagnetisation data. Scale (between ticks) of Zijderveld plot indicated in brackets. (a) Specimen SA2.3 (grey-green mudstone, Williton Member, S_1 class, polarity N); thermally demagnetised to 350 °C, followed by AF demagnetisation to 80 mT. The Zijderveld plot shows a weak northerly, Brunhes-age, magnetisation, removed at 10 mT, and a north–northeasterly-directed ChRM. (b) Specimen SA67.1 (green mudstone, Rydon Member, S_1-class, polarity R); shows the removal of a steep positive magnetisation at 350 °C, allowing isolation of a reversed ChRM with further AF demagnetisation. (c) Specimen SA66.3 (green mudstone, Rydon Member, class T_1, polarity R); thermally demagnetised to 400 °C, followed by AF demagnetisation to 80 mT. It exhibits a shallow negative SSW magnetisation, which shows erratic movement to a possibly steeper, SW-directed reverse magnetisation at high AF fields. Fitted great circle plane shown between NRM and 50 mT. (d) Specimen SA73.1 (mudstone, pre-planorbis beds; class T_2, polarity R?); displays removal of a steep northerly directed magnetisation up to about 350 °C, followed by partial great circle motion to the SW until demagnetisation instability at about 60 mT. Fitted great circle plane from NRM to 50mT steps shown. (e) Specimen SA63.3 (grey mudstone, Rydon Member; class T_1, polarity R?); displays removal of a steep positive magnetisation below 250 °C, after which a somewhat erratic, but consistently negative inclination, indicates a reverse polarity. Fitted great circle plane between NRM and the origin is shown. (f) Specimen SA34.2 (bioclastic limestone, Westbury Formation; class T_3, R?); displays great circle motion towards the SW after thermal demagnetisation at 350 °C. Fitted great circle plane between NRM and the origin is shown. Fitted upper hemisphere great circle planes shown as dotted lines, lower hemisphere as dashed lines.

specimens, in which case the length of the great-circle path is shorter. Specimens with T-type behaviour were classified qualitatively into best quality (T_1: SA66.3, Fig. 6c and e) and inferior quality (T_2, T_3: Fig. 6d and f) great-circle trends, based on the directional scatter and the degree of

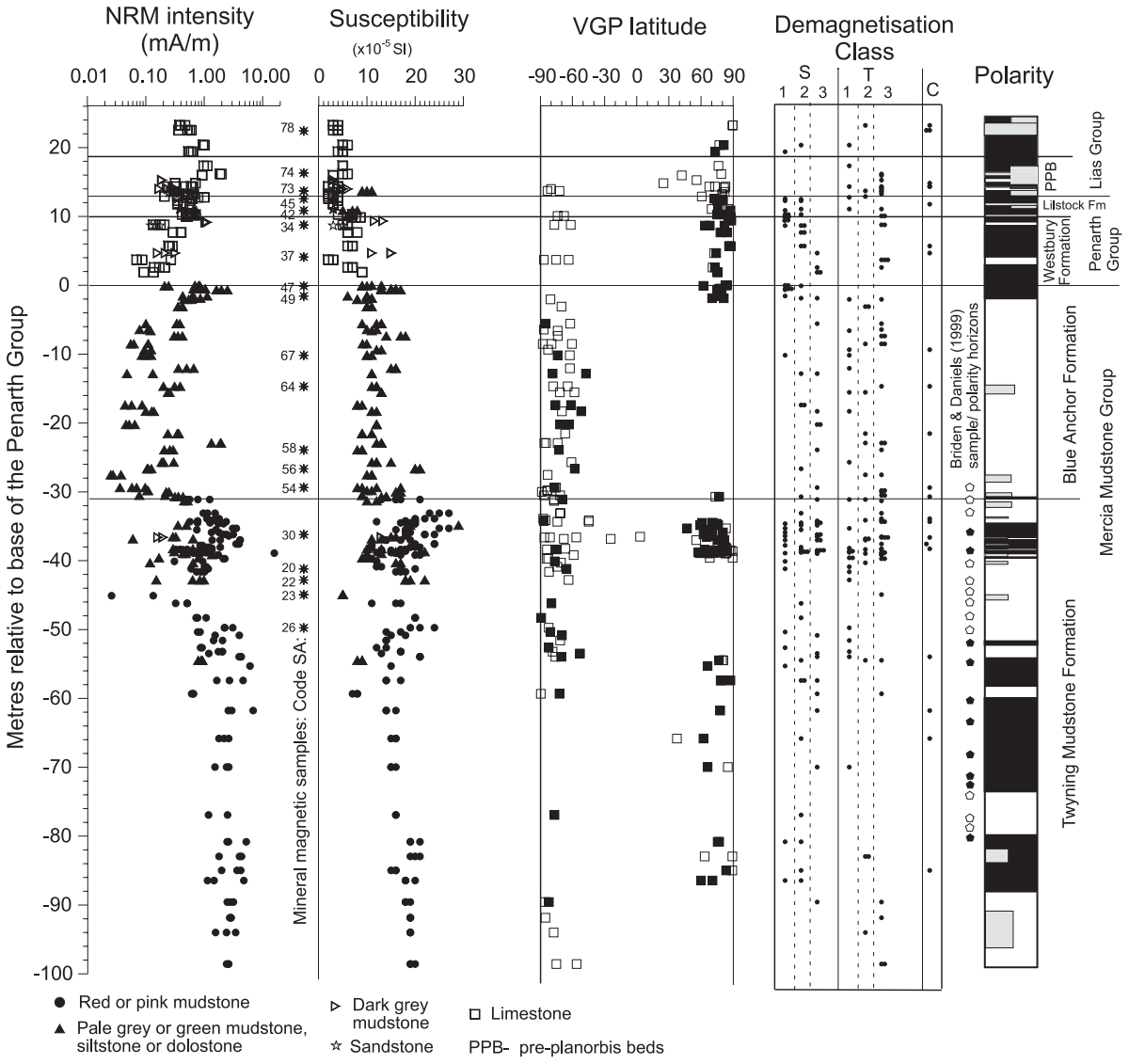


Fig. 7. Summary of the magnetostratigraphic data from the St. Audrie's Bay section. NRM intensity and volume susceptibility data plotted with symbols indicating lithology. Virtual geomagnetic pole (VGP) latitudes derived from the ChRM (filled square, S-class data) or point nearest the unit combined mean (Table 1) on the great circle trajectory (unfilled square) for each specimen. VGP latitudes approaching +90° represent normal polarity and those approaching −90° represent reversed polarity. See text for details of demagnetisation classes; each point represents a measured specimen. Specimens from horizons that were used for the magnetic mineralogy study (*) and horizon number (all code SA) adjacent to the susceptibility data. PPB, pre-planorbis beds.

approach to either SW negative or NW positive directions. T-type behaviour is seen in 42% of measured specimens (Fig. 7). It is possible to define a polarity for many T-type specimens, by the directional trend of the demagnetisation path (e.g., Fig. 6d and f).

(c) **C-Type**: Specimens exhibited either (i) erratic directional behaviour, (ii) incomplete demagnetisation (i.e., J/J_o>40%) before directional instability sets-in or (iii) the persistence of the interpreted Brunhes-age magnetisation to the highest demagnetisation stages.

In addition to line-fits evaluated for S-Type specimens, great-circle paths were evaluated for the 84% of specimens that gave a significant plane using the LINEFIND planarity test (Kent et al., 1983). Of these, 73% included the origin; the remainder gave non-origin great circle fits, mostly because of the acquisition of spurious magnetisations at high AF or thermal treatments. The average ρ was 1.8, indicating that the variance required for fitting the great circle planes is on average slightly greater than the magnetometer measured variance, and about the same as the average ρ for the line-fits.

The S quality ChRM directions (Fig. 8) were used initially to calculate Fisher mean directions (Fisher et al., 1987; Table 1). Exclusion of the slightly poorer quality S_3 data (20% of S-Type data) gave essentially the same direction, although the lower number of specimens resulted in a larger α_{95}, so the use of all S-type data was preferred. The mean for the combined Penarth and Lias groups (Fig. 8a) is steeper and more northerly directed than that from the underlying Mercia Mudstone Group (Table 1). This may be attributable to a stronger contamination of the ChRM directions in the Penarth and Lias groups by the Brunhes-age magnetisation. Lines of evidence supporting this are: (a) most of the specimens that contain a steeper (Brunhes-like) magnetisation are from the Penarth and Lias groups, and (b) the few reversed polarity samples above the Mercia Mudstone Group display southerly directed great-circle trends (T-class) that indicate significant

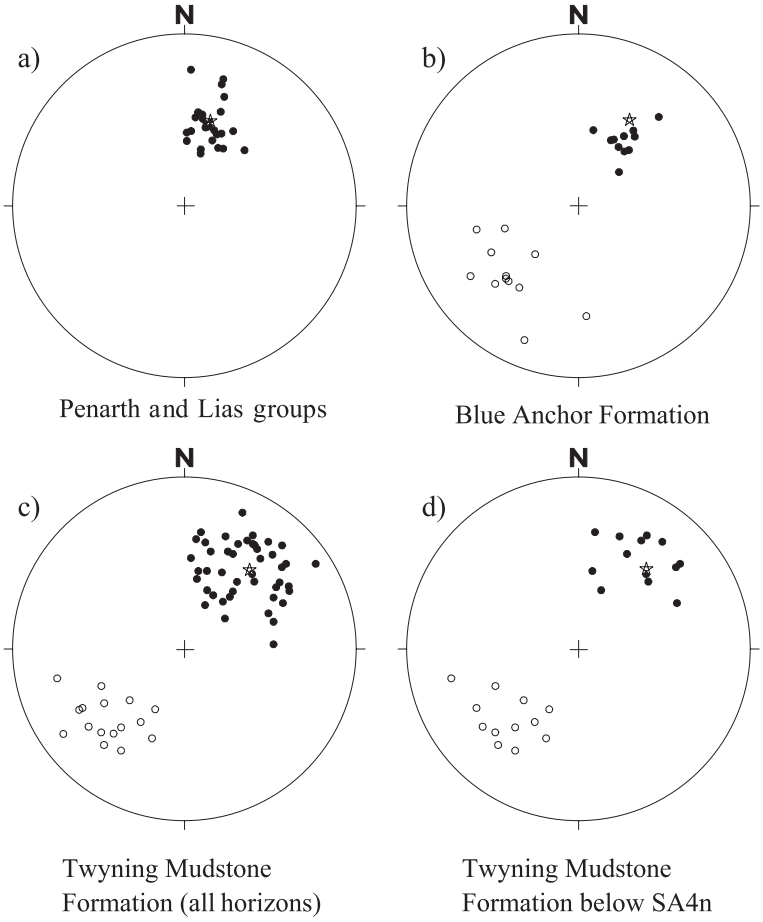


Fig. 8. Stereoplots of ChRM directions (S_1, S_2 class data) for: (a) the Penarth and Lias groups combined, (b) the Blue Anchor Formation, (c) all Twyning Mudstone Formation samples, and (d) samples from the Twyning Mudstone Formation below SA4n. The star marks the Fisher mean direction.

Table 1
Fisher mean directions for specimens with line-fits, and using combined line-fits and great circles (McFadden and McElhinney, 1988)

	$N/N_l/N_p$	Dec. (°)	Inc. (°)	k	α_{95} (°)	Reversal Test (γ°)	VGP		
							Lat. °N	Long. °E	dp/dm
Penarth and Lias groups									
Line fits	57/22/0	17.2	48.2	35.36	5.4	–	64.5	139.9	4.6/7.1
GC Combined	57/22/27	16.9	47.2	19.3	5.2	Rc (11)	63.9	141.2	4.0/6.2
Blue Anchor Formation									
Line fits	60/23/0	40.4	46.4	22.64	6.5	Rc (13)	51.7	108.9	5.4/8.3
GC Combined	60/23/28	30.6	41.5	13.47	5.7	R (14)	54.0	124.4	4.2/7.0
Twyning Mudstone Formation (all)									
Line fits	125/61/0	40.2	39.9	18.07	4.4	Rb (8)	47.9	114	3.2/5.3
GC Combined	125/61/45	39.4	40.2	16.2	3.5	Rb (6)	48.5	114.7	2.5/4.2
Twyning Mudstone Formation below SA4n									
Line fits	50/25/0	39.6	38.3	22.24	6.4	Rb (9)	47.3	115.6	4.5/7.6
GC Combined	50/25/21	40.2	39.0	17.64	5.2	Ra (5)	47.5	114.5	3.7/6.2

$N/N_l/N_p$: number of specimens measured/number with acceptable (i.e., S-class) line-fits/number with plane fits, used in calculating the mean; α_{95}: radius of the 95% cone of confidence; k: Fisher precision parameter; γ: the angle between the mean of the reverse and normal specimens. St. Audrie's Bay location 51.2°N, 356.7°E.

component overlap. To test if the directions might be contaminated, a mean was calculated using only origin intersection great circles. This used both the R and N polarity specimen planes, with the eigenvector mean (Fisher et al., 1987) being the direction common to all planes (Van der Voo, 1993). This gives a mean for the combined Penarth and Lias groups at 013°, +48°, which is not significantly different from that based on the linear-origin ChRM fits, indicating that magnetic component contamination is not apparently distorting the mean direction. Finally, the combined great circle and line-fit mean direction was calculated using the method of McFadden and McElhinney (1988), which utilises the data set more fully than line-fits alone (Table 1). Tests using both origin-included and non-origin-included plane-fits did not significantly change the mean direction; hence, all planes were used in the final combined mean directions.

The formation means pass the reversal test of McFadden and McElhinny (1990), except for the combined mean of the Blue Anchor Formation (Table 1, Fig. 8b). This is attributed mainly to the ChRM directions from the Williton Member being generally closer to those of the Penarth and Lias groups (steeper inclination, more westerly directed) than those of the underlying Rydon Member. This may be a reflection of the time gap represented by the disconformity between the Rydon and Williton members (see Section 2.1). A fold test is not possible because the strata at St. Audrie's Bay dip westwards at 8° to 15°.

The combined mean for all the Twyning Mudstone Formation samples (Fig. 8c) is slightly, but significantly different (Table 1, Fig. 9) from that given by Briden and Daniels (1999). This may be because we have a greater concentration of sampled horizons in the higher part of the Twyning Mudstone Formation (i.e., between 31 and 42 m) than Briden and Daniels (Fig. 7). Using our data from specimens below the 39 m level gives a mean that is not very different from that of the whole Twyning Mudstone Formation (Table 1), but the slightly larger 95% confidence ellipse overlaps the mean direction of Briden and Daniels (1999).

4.3. Specimen and horizon polarity

The line-fit ChRM directions (S-Class) were converted to virtual geomagnetic pole (VGP) latitude (Opdyke and Channell, 1996: p. 91) using the great circle combined mean direction of the formation, from which each specimen was obtained (Table 1, Fig. 7). For those specimens which had no line-fit, the point on their great circle trend nearest the combined mean (i.e., that used in the combined mean) was used for calculating the VGP latitude displayed in Fig. 7.

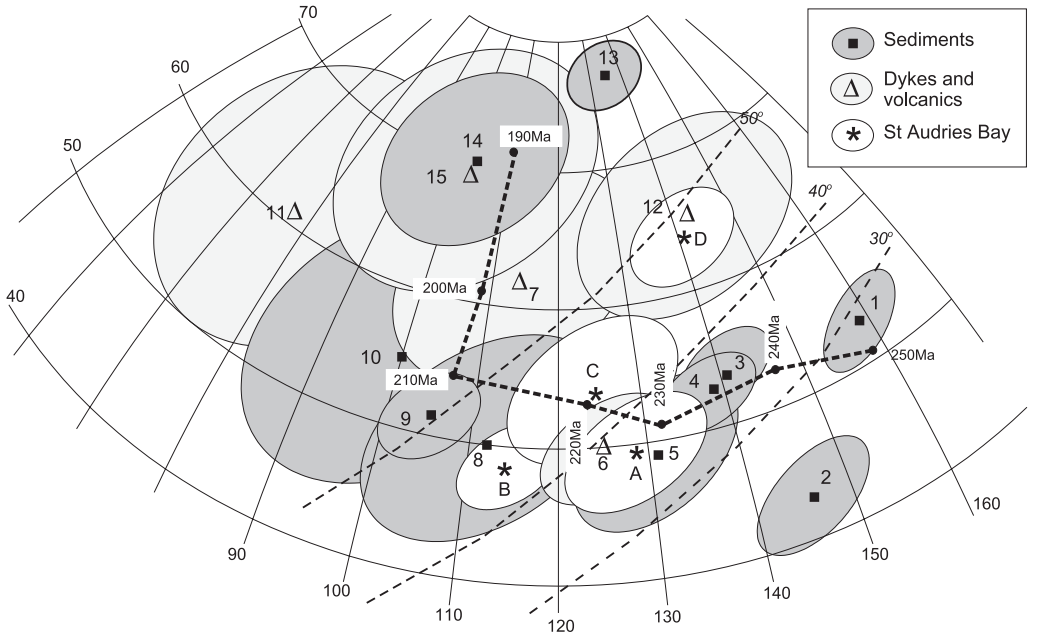


Fig. 9. Palaeomagnetic poles derived from the St. Audrie's Bay section in comparison with other European virtual geomagnetic pole (VGP) data (listed in Torsvik et al., 2001). A=Twyning Mudstone Formation (Briden and Daniels, 1999); B=Twyning Mudstone Formation (this study); C=Blue Anchor Formation (this study); D=Penarth and Lias Groups (this study). Thick dashed line: mean apparent polar wander path for Europe from Torsvik et al. (2001), at 10 Ma intervals. Thin dashed lines: isolines of inclination for the location of Bruxelles, using the European 210 Ma mean pole of Torsvik et al. (2001). VGP poles numbered: (1) Upper Buntsandstein, France; (2) Vetluga Red Clays, Russia; (3) Heming Fm (Ladinian), Paris Basin; (4) Sherwood Sandstone, Anisian, UK, Hounslow and McIntosh (2003); (5) Gipskeuper, Germany; (6) Sunhordland dykes (225 Ma); (7) Keuper volcanics (216 Ma), France; (8) 'Rhaetic' sediments, Germany and France; (9) Hettangian–Sinemurian, NE France; (10) Hettangian–Sinemurian, Paris Basin; (11) Kerforne dykes, Brittany (198 Ma); (12) Liassic volcanics (198 Ma), France; (13) Pliensbachian sediments, Yorkshire, UK; (14) Thouars and Airvault (Toarcian), France; (15) Scania basalts (179 Ma).

Specimens were also assigned a 'polarity quality' based on the quality of demagnetisation behaviour and, if from T-class specimens, the length and end-point position of the great circle trend. This was used in evaluating the horizon polarity shown in Figs. 2 and 3. One specimen of good quality polarity (S-Type, or T1-type) was sufficient to define the horizon polarity, whereas with specimens of poorer quality at least two were needed.

Major magnetozone reverse and normal couplets in the St. Audrie's section have been numbered (Figs. 2 and 3) from the base of the section, in the manner proposed by Kent et al. (1995), but using the prefix SA. This hierarchical scheme is similar to that used in numbering ocean floor anomalies, but the oldest part of a magnetozone couplet is of normal polarity (Opdyke and Channell, 1996). Seven magnetozones (SA3n.1r, SA3r.1n, SA3r.2n, SA4r.1n, SA5n.1r, SA5n.2r and SA5n.3r) are defined by multiple specimens at a single horizon. Sampling horizons adjacent to SA3r.2n (two specimens) and SA4r.1n (two specimens) indicate that these magnetozones are no more than 0.5- or 0.6 -m thick, respectively. The thicknesses of other single-horizon magnetozones are not so tightly constrained by sampling. Magnetozones SA3r.3r, SA4n.1r, SA4n.2r and SA5r are defined by sampling at two adjacent horizons, so for each, the limits imposed by the sampling indicates these magnetozones have maximum and minimum possible thicknesses of 0.6 and 0.1 m, 0.2 and 0.1 m, 0.5 and 0.3 m, and 0.6 and 0.3 m, respectively (Figs. 2 and 3).

4.4. Magnetic mineralogy

Down-section variations of NRM intensities and susceptibilities largely reflect changes in lithology (Fig. 7). The highest NRM intensity and susceptibilities are seen in the red mudstones in the Twyning

Mudstone Formation. Within the Blue Anchor Formation, the largest NRM intensities correspond to a yellowish dolostone horizon in the Rydon Member and grey mudstone horizons in the Williton Member. Specimens from the Penarth and Lias groups have similar NRM intensities to those from the Blue Anchor Formation, but because they are dominantly from micritic silty limestones, they show the lowest susceptibilities. Hysteresis measurements (up to 1 T) indicate that the non-red lithologies are dominated by a paramagnetic behaviour (consistent with magnetic torque measurements of Hounslow, 1985), whereas the red mudstones have, in addition, a nonsaturating high coercivity (>100mT) component, consistent with haematite being present.

IRM data show that the specimens are largely divisible into those, like SA74.1, SA78.1a and SA34.3, which are dominated by low coercivity minerals (magnetite or magnetic sulphides); and those such as SA22.2, which are dominated by high coercivity minerals (haematite or goethite; Fig. 10a). The latter are mostly red-coloured, although some greenish mudstones may also acquire substantial IRM above 300 mT. Some specimens from the lower parts of the Rydon Member (SA58.1) and from the pre-planorbis beds of the Lias Group (SA73.3) have a mixture of both magnetite-like and haematite (or goethite)-like IRM components (Fig. 10a).

Orthogonal IRM demagnetisation of red mudstones (e.g., SA22.3, Fig. 10b) indicates a maximum blocking temperature of 700 °C and confirms that haematite is the dominant high coercivity material, although a pale reddish-grey mudstone (horizon SA23, Fig. 7) has a low blocking temperature (<150 °C) mineral (goethite?) and a magnetite-like blocking temperature. There may also be evidence of goethite in other magnetite-dominated specimens, such as from horizon SA47 (Williton Member) and SA58.2A from low in the Rydon Member (Figs. 7 and 10b). Thermally distributed blocking temperatures up to ~550 °C (e.g., SA58.2a, Fig. 10b) are typical of other

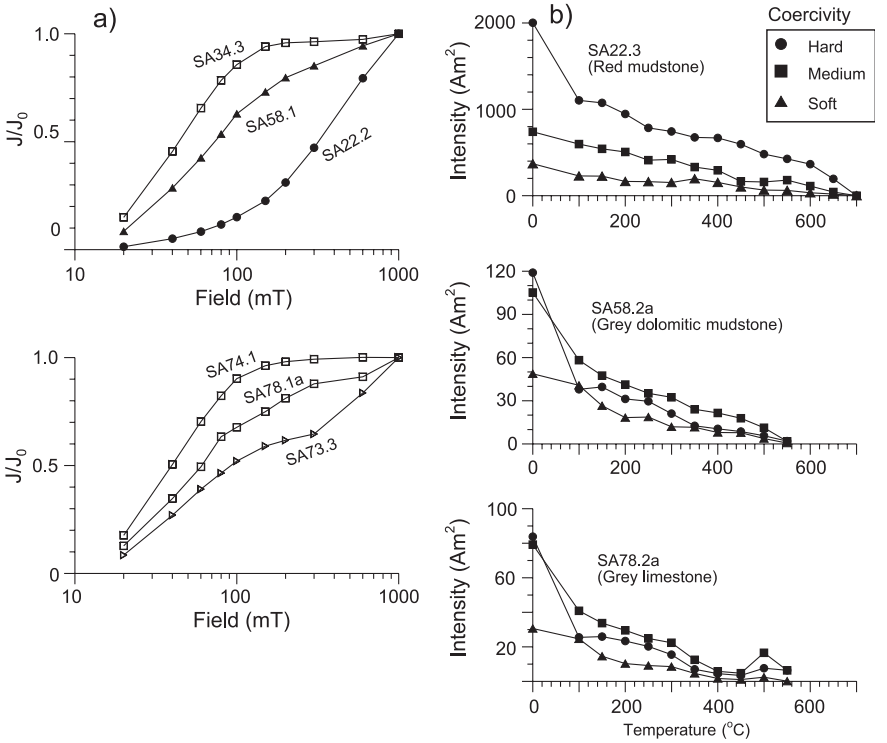


Fig. 10. (a) Isothermal remanent magnetisation (IRM) acquisition curves for a subset of representative specimens. Symbols indicate lithology, as in Fig. 7. (b) Thermal demagnetisation of three-component orthogonal IRM for representative specimens.

magnetite-dominated specimens. Specimen SA78.2a, from the Lias Group, shows evidence of a possible magnetic sulphide (Fig. 10b), by more rapid decrease of all coercivity fractions between 300 and 400 °C (Roberts et al., 1999). Loss of both high and moderate coercivity components at 100 °C (e.g., SA78.2a; Fig. 10b) may be due to oxidation of this sulphide on thermal demagnetisation.

In conclusion, the mineralogical studies indicate that the magnetisation of red mudstones in the Twyning Mudstone Formation is dominantly carried by haematite (in some cases with goethite), with both reversed and normal polarities. The Blue Anchor Formation and the Penarth and Lias groups are dominated by a magnetite-like phase, which again carries both polarities. There is weak evidence, from the three-component IRM, that a magnetic sulphide may be important in some Lias Group samples, and possibly also in some from the Penarth Group. This magnetic sulphide may be partly responsible for the stronger Brunhes-age overprint in those samples.

5. Discussion

5.1. Palaeomagnetic poles

The VGPs from St. Audrie's Bay appear to lie in the transition from the west-directed, European Permian–Triassic apparent polar wander (APW) track to the north–east-directed Jurassic–Cretaceous APW track (Fig. 9). The stratigraphic order of VGPs in the St. Audrie's Bay section appears to show principally the Jurassic, north–east-directed, motion. As noted by Torsvik et al. (2001), the Lower to Middle Jurassic VGP data from cratonic Europe are sparse, with more dispersed VGPs than those from the Permian and Triassic. There is less dispersion if poles based on sediments and those based on volcanic successions are considered separately; the Lower Jurassic volcanic-based VGPs, giving more pole-ward displacement (steeper apparent inclination) than those based on sediments (Fig. 9). This could be inferred to indicate 10°–15° inclination shallowing in the sediments. However, Van der Voo and Torsvik (2001) discounted inclination shallowing as a viable explanation for discrepancies in the complete Pangea data set, but the small Upper Triassic and Lower Jurassic dataset is suggestive of this possibility.

5.2. Magnetostratigraphy

The magnetostratigraphy reported here amplifies and considerably extends that obtained by Briden and Daniels (1999). The additional sampling largely substantiates Briden and Daniels' results from the Twyning Mudstone Formation (Figs. 2 and 9). However, a minor reversal (SA3n.1r) is indicated by one sampling horizon, between Briden and Daniels' horizons 14 and 15, and other samples from the present study indicate reversed polarity between their horizons 13 and 14 (Fig. 2). Briden and Daniels' horizon 13 thus appears to be anomalous or comprises a short-duration normal magnetozone (SA3r.1n) in SA3r.

The disconformity between the Rydon and Williton members of the Blue Anchor Formation (Section 2.1) coincides with a change in polarity, from SA4r to SA5n. Mayall (1981) proposed that there is a regional disconformity at the boundary between the Blue Anchor and Westbury formations, and that equivalents of the Williton Member are missing or condensed at localities in South Wales and south Devon. Hence, the normal polarity corresponding to the Williton Member may comprise either part of the overlying normal polarity, or a different normal magnetozone. The Penarth Group appears to encompass three reverse magnetozones; these are defined by samples from single horizons and therefore could be regarded as tentative, until confirmed from other sections. However, support for reversals at about this level is provided by the existence of reverse polarity magnetisations in the lithostratigraphic equivalents of the Penarth Group in western Germany and north-eastern France (Edel and Duringer, 1997), and magnetic polarity data from the Penarth Group at Lavernock Point in South Wales (Hounslow et al., 2002).

5.3. Comparison with the Newark Supergroup, eastern USA

Of the published Upper Triassic magnetostratigraphic records, that from the Newark Supergroup in

eastern North America (Kent et al., 1995) is the most readily comparable with that from St. Audrie's Bay, in terms of both the order and the relative thickness of the magnetozones recorded (Fig. 11). The features in common between the St. Audrie's Bay succession and the biostratigraphy in the Newark Supergroup are:

(A) They have a few miospore taxa in common, these being *Granuloperculatipollis rudis*, *Alisporites thomasii*, *Tsugaepollenites? pseudomassulae*, *Carnisporites spiniger*, *Carnisporites leviornatus* and *Porcellispora longdonensis*. Of these, *T. pseudomassulae* and *P. longdonensis* have both first occurrence (FAD) and last occurrence (LAD) close to the boundary of the Upper Balls Bluff–Upper Passaic and *Corollina meyeriana* palynofloral zones (Cornet, 1993; Fowell and Olsen, 1993; Fig. 11). *C. spiniger* also has a consistent LAD at or just above the base of the *C. meyeriana* palynofloral zone. *A. thomasii* has a FAD near the Newark Supergroup Triassic–Jurassic palynofloral boundary (Cornet, 1993; Fowell et al., 1994; Fig. 11). The LAD of *G. rudis* occurs in the upper part of the Upper Balls Bluff–Upper Passaic palynofloral zone and its FAD is in the upper part of the Lower Passaic–Heidlersberg palynofloral zone (Cornet, 1993; Fowell and Olsen, 1993). *C. leviornatus* is found in the transition interval between the Upper Balls Bluff–Upper Passaic palynofloral zone and the underlying Lower Passaic–Heidlersberg palynofloral zone according to Fowell and Olsen, (1993) and throughout the Upper Balls Bluff–Upper Passaic palynofloral zone according to Cornet (1993).

(B) A change in the terrestrial microflora that occurs in the upper part of the Penarth Group at St. Audrie's Bay (and elsewhere in the UK at a comparable horizon) is similar to that interpreted as marking the Triassic–Jurassic boundary in the Newark Supergroup (Fowell et al., 1994). In both cases, an upward increase in trilete spore diversity and abundance (although with different species) is followed by a rapid decline and a change to low diversity assemblages dominated by circumpolles. Hence, these events appear to represent comparable microfloral turnovers.

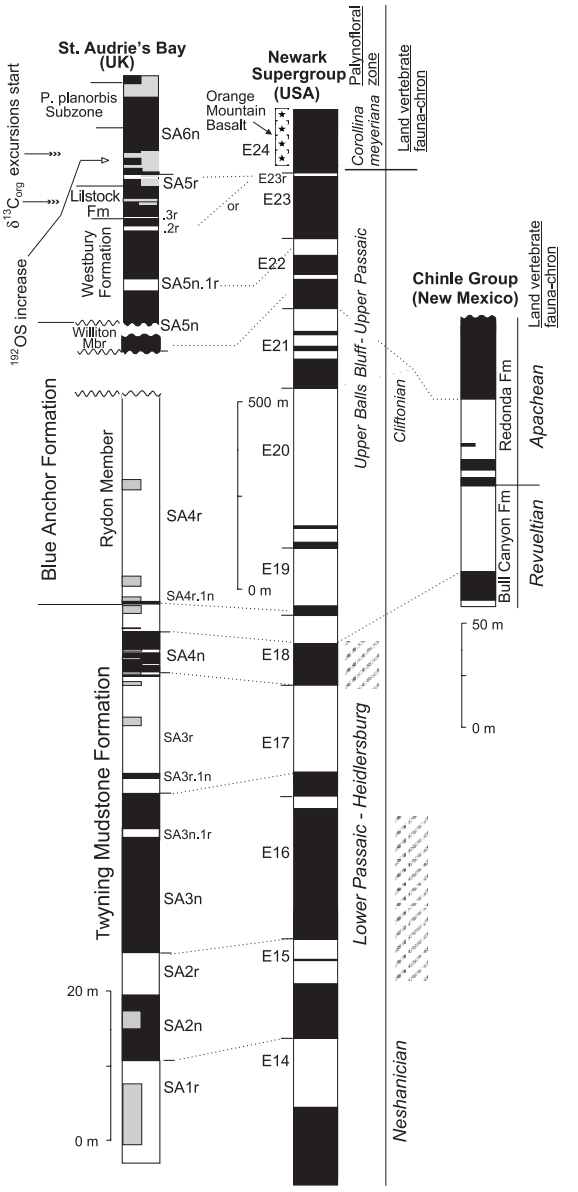


Fig. 11. Proposed magnetostratigraphic correlation between the St. Audrie's Bay section with nonmarine sections of comparable age. Newark Supergroup details from Kent et al. (1995) and Kent and Olsen (1999). Chinle Group magnetostratigraphy from Molina-Garza et al. (1996) and Reeve and Helsley (1972); land vertebrate faunachron divisions from Lucas (1999). Individual metre scales for each section. The hatched areas indicate the uncertainty in the placement of biostratigraphic zonal boundaries. $\delta^{13}C_{org}$ excursions from Hesselbo et al., (2002).

(C) Lucas (1999) suggests the Revueltian and Apachean land vertebrate faunal zones are equivalent to the eastern North American Cliftonian and Neshancian land vertebrate faunal zones. Vertebrates from fissures in Carboniferous limestone in the Bristol region (UK) include elements indicative of the Revueltian and Apachean land vertebrate faunal zones (Lucas, 1998, 1999). The age of the fissure fills is not precisely known, but some have faunal elements in common with those in the Westbury Formation (Storrs, 1994; Swift and Martill, 1999), and the dinoflagellate cyst *Rhaetogonyaulax rhaetica* also occurs in some (Marshall and Whiteside, 1980).

Biostratigraphic comparison therefore indicates that the palynologically productive parts of the St. Audrie's Bay section may equate approximately with the Upper Balls Bluff–Upper Passaic palynofloral zone through to the Triassic–Jurassic transition interval in the Newark Supergroup.

Correlation of the magnetostratigraphy of the Newark Supergroup with that at St. Audrie's Bay is primarily based upon the identification of the equivalents of SA3n and SA4r magnetozones (Fig. 11). SA2n and SA2r correspond with the E15, normal-reversed couplet. SA3n is correlated with the E16n to E17n interval and includes the magnetozone SA3n.1r which appears to best equate with E16r; SA3r is correlated with E17r and SA4n with E18n. The magnetozone SA4r.1n, in the basal part of the Blue Anchor Formation, probably corresponds with E19n. The two normal intervals in the lower part of E20 have not been detected in SA4r at St. Audrie's Bay, probably because of the wide sample spacing in the lower part of the Rydon Member where they would be expected to occur (Fig. 2). In addition, several short-duration reverse magnetozones in SA4n, which are in part represented by sampling at adjacent horizons, have not been detected in E18n.

The normal polarity interval SA5n to SA6n continues to the top of the sequence studied at St. Audrie's Bay (Fig. 3). This dominantly normal polarity appears generally consistent with magnetostratigraphic studies of the Hettangian in the Paris Basin and Germany (Yang et al., 1996; Edel and Duringer, 1997). The minor reverse magnetozone SA5r, defined by three specimens on two horizons, is between 0.28- and 0.59-m thick; its lower boundary is 0.6 m above the base of the Lias Group and its upper boundary is 4.6 m below the level of the appearance of definite *Psiloceras planorbis*. A reversed interval about 2 m above the top of the 'Rhaetic' in Xeuilley quarry, northeastern France (Edel and Duringer, 1997), may be equivalent to this magnetozone, but there, its relationship to the level of the appearance of *Psiloceras* is not clear.

5.4. The Triassic–Jurassic transition

In the Newark Supergroup, the change in terrestrial palynofloras that has been taken as marking the Triassic–Jurassic boundary (Fowell and Olsen, 1993; Fowell et al., 1994; Fowell and Traverse, 1995) is similar in style to that seen at St. Audrie's Bay (and other UK sections) within the Lilstock Formation. This marked change to low-diversity assemblages dominated by circumpolles is also a distinctive feature of successions in southern Sweden and Denmark (Batten and Koppelhus, 1996; Koppelhus and Batten, 1996), in the Alps (Schuurman, 1979) and in the Germanic-type facies in France, Germany and Poland (Schuurman, 1977, 1979). A major turnover from ornamented trilete spores to smooth-walled species also defines the MD to AM biozone boundary of Suneby and Hills (1988) in the Sverdrup Basin. It could be assumed that all these locations are displaying synchronous terrestrial palynofloral changes linked to the environmental and biotic changes in the Triassic–Jurassic boundary interval. However, this assumption is questioned using the magnetostratigraphic data.

Alternatively, Hesselbo et al. (2002) have suggested that the major palynofloral changes occur lower, within the Cotham Member, where an initial organic carbon isotopic anomaly is detected. This level approximately corresponds to the phase 3 to 4 transition of Schuurman (1979), where *Ovalipollis pseudoalatus*, *Granuloperculatipollis rudis* and *Rhaetipollis germanicus* become scarce and *Kraeuselisporites reissingeri* becomes consistently present. Whilst these features are present in the Penarth Group and other Germanic type facies (Schuurman, 1977), palynofloral zonal schemes in other areas do not utilise these forms consistently (Morbey, 1975; Suneby and Hills, 1988; Koppelhus and Batten,

1996). Nor does the phase 3 to 4 transition of Schuurman (1979) maintain a consistent relationship with marine biostratigraphy. In the Kendelbach section, this same transition is located in the pre-planorbis strata, above beds containing the ammonite *Choristoceras marshi* and conodonts (Schuurman, 1979; Golebiowski, 1990a). In St. Audrie's Bay and other UK sections, this transition also occurs prior to the last occurrences of conodonts. Hence, it is difficult to support the suggestion by Hesselbo et al. (2002) that this microfloral transition is as regionally significant as that higher in the Penarth Group.

Cohen and Coe (2002) have interpreted Os and Re changes as indicating the initial weathering of basalts from the Central Atlantic Magnetic Province (CAMP) during the deposition of the pre-planorbis beds. They suggest the short residence time of Os in the oceans (~40 kyr) indicates Os deposition would be effectively synchronous with its weathering input into the Triassic oceans. Significant Os increase begins at a level between 2.6 and 3.3 m below the Planorbis Zone, that is c. 1.6 m above the top of SA5r. This increase is coincident with the main organic carbon isotope excursion in the St. Audrie's Bay section (Figs. 3 and 11) that Hesselbo et al. (2002) have interpreted as due to volcanism. These data suggest that the sequence of environmental events seen at St. Audrie's Bay and in the Newark Supergroup is closely comparable.

However, the magnetostratigraphic evidence conflicts with this simple sequence of events in that SA5r occurs between the palynofloral turnover and Os increase, whereas the reverse magnetozone E23r occurs prior to the palynofloral turnover and flood basalts in the Newark Basin. There are two plausible explanations:

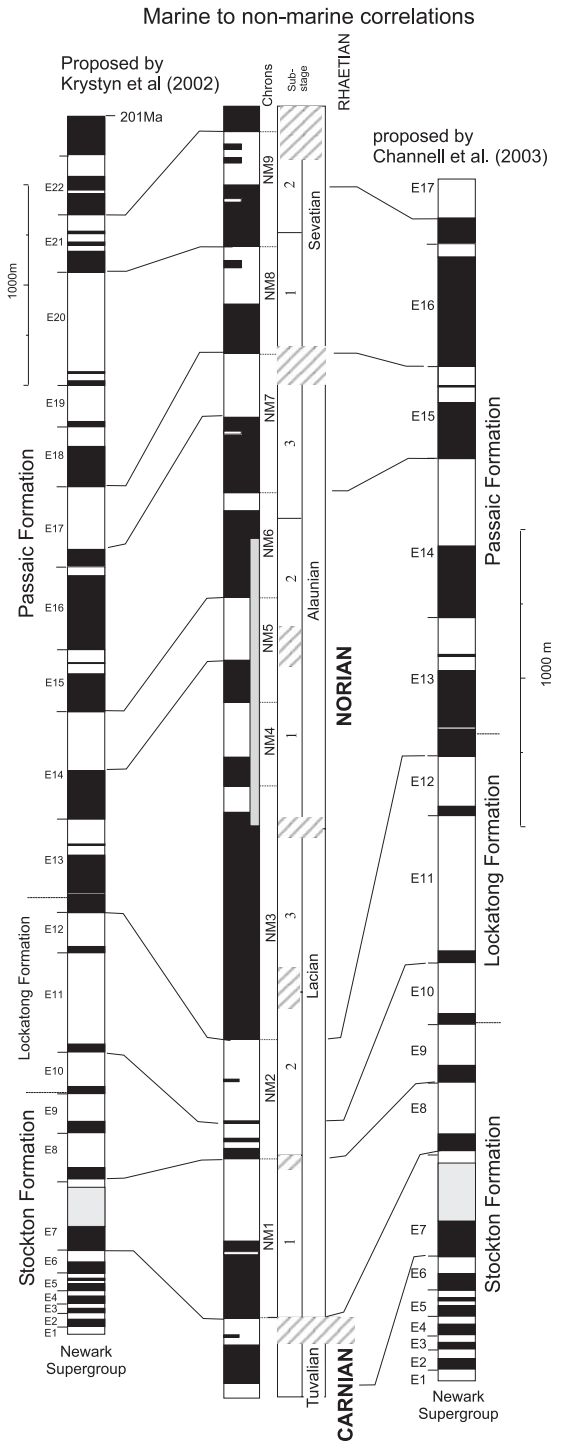
(a) The reverse magnetozone SA5r is not the equivalent to E23r; instead SA5n.2r, or SA5n.3r may be the equivalent of E23r and the correlative of SA5r is currently undetected within the Newark Basin flood basalt succession. A flood basalt succession from Morocco dated to the Triassic–Jurassic transition, includes a short reverse polarity interval and interbedded sediments containing Upper Triassic miospores (Marzoli et al., in press), which seems to support this possibility. This would suggest there might be two (or more) short duration reverse polarity intervals, one predating the microfloral turnover (e.g., SA5n.2r/SA5n.3r and E23r and Moroccan reverse) and a later one shortly after the turnover (i.e., SA5r). One implication of this possibility is that the base of the *Corollina meyeriana* palynofloral Zone in the western USA correlates with Triassic conodont-bearing strata of uppermost Rhaetian age in the UK.

(b) The microfloral turnover in the St. Audrie's Bay succession and the Newark Supergroup was not synchronous, but occurred later in the eastern USA, than in the UK. This is supported to some extent by the occurrence of a microfloral turnover, comparable to that near the top of the Lilstock Formation, within, or at the base of the Planorbis Zone in the Northern Calcareous Alps (Schuurman, 1979). Such microfloral changes might not be synchronous if they are related to terrestrial palaeogeographic and environmental changes caused by sea level change, such as that reflected in the European stratigraphic record through the Rhaetian and the Triassic–Jurassic boundary interval.

5.5. Correlation to the marine Upper Triassic

The St. Audrie's Bay magnetostratigraphy has a much better match with the Newark Supergroup magnetostratigraphy than with the marine Norian equal biozone composites of Gallet et al. (2000) and Krystyn et al. (2002). Consequently, resolution of the magnetostratigraphic correlation of Norian marine and nonmarine successions is not immediately improved, relative to previous contributions (Kent et al., 1995; Muttoni et al., 2001; Channell et al., 2003; Gallet et al., 2003). A number of different suggestions for correlating mid-Norian and younger marine sections with the Newark Supergroup have been proposed (Fig. 12).

The correlations suggested by Channell et al. (2003) places Sevatian strata equivalent to the E16 and E17r magnetozones of the Newark Supergroup (Fig. 12). However, this leads to E13n and E14n having four potentially equivalent normal magnetozones (MN3n, MN4n, MN5n and MN6n) in the Krystyn et al. (2002) composite (Fig. 12) and



implies that the Newark Supergroup magnetostratigraphy is not a complete record of the magnetic field polarity.

Krystyn et al. (2002) and Gallet et al. (2003) suggest an alternative plausible correlation of the polarity pattern of the Sevatian with the E18n to E22n interval of the Newark Supergroup (Fig. 12). This correlation suggests that in terms of time as measured in the cyclostratigraphy of the Newark Supergroup, NM8r has a time duration some three times longer than NM7r. However, this is inconsistent with the relative thickness in metres of these reverse magnetozones in the Kavur Tepe and Silická Brezová sections (Gallet et al., 1993, 2000; Channell et al., 2003). In both these sections, NM8r is shorter by 36% and 35%, respectively (in metres), than the equivalent NM7r. It will be shown instead, that NM7r is most likely to correlate to the E18r to E20r interval of the Newark Supergroup, which produces a more favourable comparison in relative thickness of these magnetozones, between the Newark Supergroup and the Kavur Tepe and Silická Brezová sections (Figs. 13 and 14).

5.6. New insight using data from the Wombat Plateau

The data collected from the NW Australian margin during Ocean Drilling Project (ODP) Leg 122 provides additional insight into the correlation of the marine and nonmarine Upper Triassic, through the occurrence of the dinoflagellate cyst *Rhaetogonyaulax rhaetica*, and of the magnetostratigraphy (Fig. 13). The ODP Leg 122 cores contain the *Samaropollenites speciosus*, *Minutosaccus crenulatus* and *Ashmoripollis reducta* palynofloral zones (Brenner, 1992) which have been calibrated with a conodont biostratigraphy from the same region (Nicol and Foster, 1994). The *A. reducta* zone is constrained within the *Misikella hernsteini* and *Misikella posternsteini* conodont zones, which equate in part with the Sevatian 2 definition used by Gallet et

Fig. 12. Comparison of the Tethyan Norian composite (Krystyn et al., 2002; Gallet et al., 2003) with the Newark Supergroup magnetostratigraphy, and the correlations proposed by Channell et al. (2003) and Krystyn et al. (2002). The Stockton Formation magnetostratigraphy is modified according to Le Tourneau (1999). The hatched areas indicate the uncertainty in the placement of stage and substage boundaries in relationship to the magnetostratigraphy.

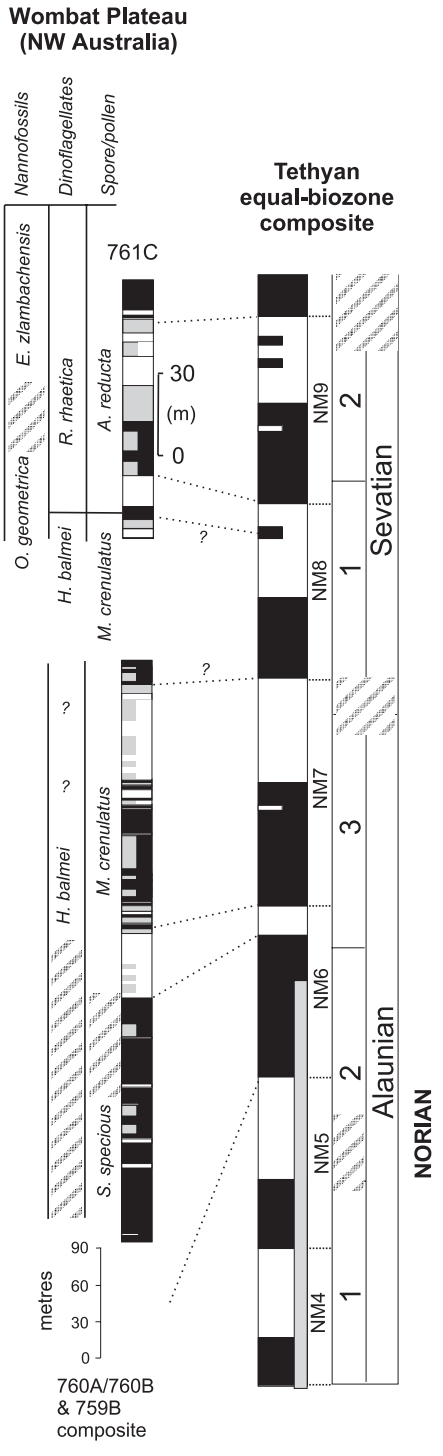


Fig. 13. Comparison of the biostratigraphy and magnetostratigraphy of the Wombat Plateau cores with the equal biozone composite of Krystyn et al. (2002). Biostratigraphy for the Wombat Plateau cores (ODP Leg 122) from Brenner (1992) and Brenner et al. (1992) and magnetostratigraphy from Galbrun (1992). The hatched areas indicate the uncertainty in the placement of stage, substage and biostratigraphic zonal boundaries.

al. (1992, 1996). The *M. crenulatus* palynofloral zone ranges from the lower Alaunian (approximately the *Epigondella multidentata* conodont zone) to the *M. hernsteini* conodont zone. The *Eoconusphaera zlambachensis* nannofossil biozone in core from ODP Site 761C indicates a possible age younger than mid-Norian (younger than the later part of the *Mesohimavatites columbianus* ammonite zone; Bown, 1998; Brenner et al., 1992; Fig. 13). The FAD of *R. rhaetica* occurs in ODP hole 761C and is within the *Sagenites reticulatus* ammonite biozone (upper Sevatian) in the Kendelbach section (Morbey, 1975, Krystyn, 1990; Golebiowski, 1990a).

Therefore, the integrated bio- and magnetostratigraphy from the ODP Leg 122 cores provides a clear match to the Tethyan composite, from the mid-Alaunian to the upper Sevatian 2 (Fig. 13). In particular, this correlation suggests that the normal polarity magnetozone in the lowest parts of the cores at ODP Sites 760 and 759 equates with magnetozone NM6n, with the remaining core from these sites showing an interval of normal and reverse polarity, similar to that within the remaining Alaunian 3 subzone (Fig. 13). The magnetostratigraphic correlation of these places the lowest likely FAD of *Rhaetogonyaulax rhaetica* in an interval within the upper part of NM8r (Figs. 13 and 14).

5.7. A bio-magnetostratigraphic scale for the Upper Triassic

It was suggested in Section 5.5 that the correlation of NM7r with the E18r to E20r interval in the Newark Supergroup is more consistent with the relative thickness in metres of NM8r and NM7r in the Kavur Tepe and Silická Brezová sections. This provides one basis for the correlations proposed in Fig. 14. Additional biostratigraphic links that support the proposed correlation of the Tethyan composite magnetostratigraphy of Gallet et al. (2000) with the

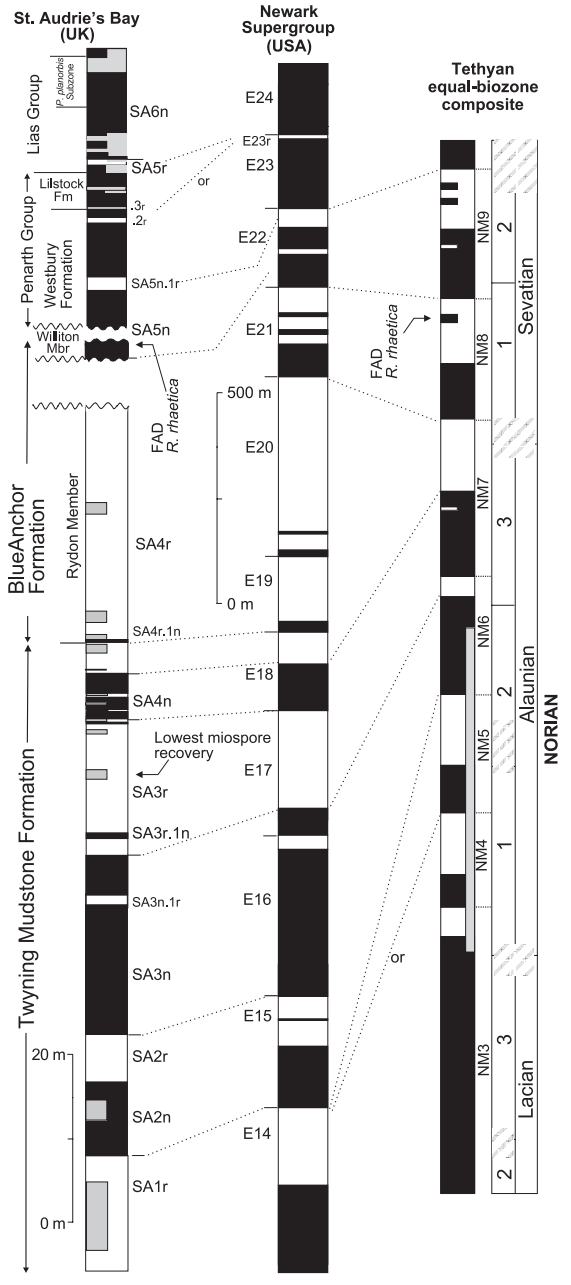


Fig. 14. Correlation proposed between the marine Norian and the magnetostratigraphies at St. Audrie's Bay and the Newark Supergroup. See text for details. The hatched areas indicate the uncertainty in the placement of stage, substage and biostratigraphic boundaries.

St. Audrie's Bay and Newark Supergroup magnetostratigraphies are:

(1) The FAD of the dinoflagellate cyst *Rhaetogonyaulax rhaetica* occurs in the Williton Member, in the base of ODP hole 761C (Fig. 13), and within the *Sagenites reticulatus* ammonite biozone (upper Sevatian) in the Kendelbach section, suggesting that the Williton Member is probably Sevatian 2 in age. This constrains the overlying Penarth Group to Sevatian 2 in age or younger. Hence, SA5n.1r could correlate with NM9r, as the most likely possibility.

(2) From the lower part of the Lilstock Formation to the lowest recovery of palynomorphs from the St. Audrie's Bay section (Figs. 4 and 14), the palynology suggests a close comparison to the phase 3 assemblage zone of Schuurman (1979). A marked upward increase in palynomorph diversity is a characteristic feature of this assemblage. Schuurman (1979) compared this assemblage with the RG, RK, LR, LL, Me and Mi zones and subzones defined by Morbey (1975) in the Kendelbach section, which there ranges in age from the *Sagenites reticulates* ammonite biozone of the Sevatian, through to the pre-planorbis interval (post-conodont) part of the Kendelbach section (Schuurman, 1979; Golebiowski, 1990a,b). A lower age boundary for this palynological assemblage is uncertain, but it extends at least into the *Mesohimavatites columbianus* ammonite biozone of the upper Alaunian, based on data from Arctic Canada (Fisher and Dunay, 1981). Thus, in the St. Audrie's Bay section, the lowest recovery of palynomorphs may be in strata as old as upper Alaunian. This is consistent with the magnetostratigraphic correlations that place the base of SA3r (and E17r) within the lower part of Alaunian 3 (Fig. 14). Correlation of the magnetozone interval SA1r–SA2n to the marine composite, over the interval Alaunian 1 to the lower part of Alaunian 2, is less certain because of the stratigraphic and tectonic complexities of the sections which make up the marine composite (Fig. 14).

(3) The Revueltian land vertebrate faunachron equates in part to the Neshanician and the lower part of the Cliftonian land vertebrate faunachron of

the Newark Supergroup (Lucas, 1998; Fig. 11). A Revueltian vertebrate fauna is known from the Calcare de Zorcino and the Forni Dolomite of the Lombardy Alps and provides independent evidence of a youngest age of mid to latest Alaunian (approximately the *Mesohimavatites columbianus* Zone) for the Revueltian land vertebrate faunachron (Lucas, 1999). Hence, the interval E18 to E20 (SA4n, SA4r) in the Newark Supergroup is probably mid to latest Alaunian at the youngest (Figs. 11 and 14). This is consistent with the palynology and magnetostratigraphy.

6. Conclusions

The magnetostratigraphy from St. Audrie's Bay, UK provides strong evidence that the Newark Supergroup magnetostratigraphy is a good representation of the Upper Triassic magnetic field over the E14r to E20r interval. The biostratigraphic and magnetostratigraphic data from St. Audrie's Bay allows some resolution of the stage and substage assignments of the Newark Supergroup magnetostratigraphic polarity pattern. These suggest that the Alaunian–Sevatian boundary occurs close to, or just below the base of the Newark Supergroup E21n magnetozone and at the disconformity between the Rydon and Williton members of the Blue Anchor Formation. The placement of the top of the Sevatian (base of the Rhaetian) is less clear, mainly because of the lack of a consensus on the biostratigraphic definition of this boundary. Using the lowest and highest likely biostratigraphic boundary definitions, it falls between the middle part of E22r and the lower part of E23n. The magnetostratigraphic correlations suggest that the lower part of the Westbury Formation (Penarth Group) and Williton Member are Sevatian in age, and the Blue Anchor Formation is late Alaunian. The age of the underlying Twyning Mudstone Formation at St. Audrie's Bay ranges down into the mid or early Alaunian (mid-Norian).

At St. Audrie's Bay, the reverse magnetozone SA5r occurs within the pre-planorbis beds and above a major microfloral turnover within the Rhaetian; whereas in the Newark Supergroup, an equivalent turnover is located above a short reversed interval (E23r). SA5r and E23r may be equivalents, indicating the microfloral turnovers are not synchronous. A similar microfloral turnover in other European sections is not consistently located in relationship to marine biostratigraphic indicators, showing that regional environmental factors probably controlled the timing of this event. An alternative possibility, not discounted by the magnetostratigraphy from St. Audrie's Bay, is that the microfloral turnovers at St. Audrie's Bay and in the Newark Supergroup were synchronous, suggesting that the SA5r reverse magnetozone is equivalent to an undetected reverse magnetozone within the Newark Basin flood basalts. This would place the initiation of the Central Atlantic Magmatic Province flood basalts within the Rhaetian. Further detailed magnetostratigraphic work on other Triassic–Jurassic boundary sections is required to fix the relative timing of environmental events, and magnetic reversal intervals at this boundary.

Acknowledgements

This work was supported by Norsk Hydro, Saga Petroleum, Deminex and a Leverhulme Trust Grant to MWH and Julian Andrews, who assisted with some fieldwork to St. Audrie's Bay. Steve Hesselbo provided useful discussion about the section. Paul Olsen and an anonymous reviewer, as well as Stewart Molyneux and Andy Howard provided helpful comments on a previous version. GW publishes with the approval of the Executive Director of the British Geological Survey (NERC). This is a contribution to IGCP project 458 "Triassic–Jurassic boundary events".

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